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Lin et al.

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(54) **SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURE**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Taiwan Semiconductor Manufacturing Company**, Hsinchu (TW)
(72) Inventors: **You-Ru Lin**, New Taipei (TW); **Sheng Kai Yeh**, Taichung (TW); **Jen-Yuan Chang**, Hsinchu (TW); **Chi-Yuan Shih**, Zhubei (TW); **Chia-Ming Hung**, Taipei (TW); **Hsiang-Fu Chen**, Zhubei (TW); **Shih-Fen Huang**, Jhubei (TW)

11,444,068 B2 * 9/2022 Song H01L 24/80
11,862,618 B2 * 1/2024 Lee H01L 23/5226
12,009,303 B2 * 6/2024 Kim H01L 23/36
12,057,383 B2 * 8/2024 Haba H01L 24/32
2008/0061419 A1 3/2008 Enquist et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 106365115 B 8/2019
TW 1560135 B 12/2016

(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

Primary Examiner — Telly D Green

(74) *Attorney, Agent, or Firm* — Lippes Mathias LLP

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(57) **ABSTRACT**

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(51) **Int. Cl.**
H01L 23/31 (2006.01)
B81C 1/00 (2006.01)

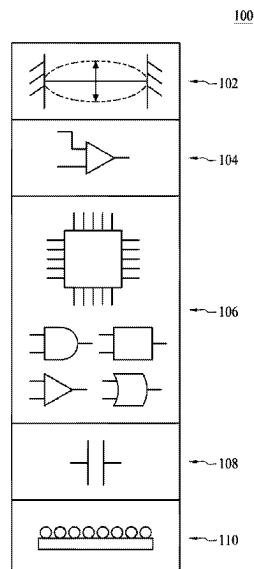
(52) **U.S. Cl.**
CPC **H01L 23/3114** (2013.01); **B81C 1/00261** (2013.01); **B81C 2203/036** (2013.01); **B81C 2203/0714** (2013.01)

(58) **Field of Classification Search**
CPC H01L 23/3114; B81C 1/00214; B81C 1/00261; B81C 2203/036; B81C 2203/0714; B81B 2201/0271; B81B 2203/0127

See application file for complete search history.

A semiconductor device and method of forming such a device includes a MEMS component including one or more MEMS pixels and having a MEMS membrane substrate and a MEMS sidewall. The semiconductor device includes an analog circuit component bonded to the MEMS component, and which includes at least one analog CMOS component within an analog circuit insulative layer, and an analog circuit component substrate. The semiconductor device includes an HPC component bonded to the analog circuit component substrate. The HPC component includes at least one HPC metal component disposed within an HPC insulative layer, at least one bond pad, at least one bond pad via connecting the at least one bond pad and the at least one HPC metal component, and an HPC substrate. Additionally, the semiconductor device includes a DTC component bonded to the HPC substrate, and which includes a DTC die disposed in a DTC substrate.

20 Claims, 26 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0081458	A1 *	4/2008	Lin	H01L 24/11 438/620
2010/0093259	A1 *	4/2010	Lee	B24B 49/12 451/6
2013/0168740	A1	7/2013	Chen	
2017/0053907	A1	2/2017	Huang et al.	
2021/0155472	A1	5/2021	Tocchio et al.	
2021/0351135	A1	11/2021	Or-Bach et al.	
2022/0020735	A1 *	1/2022	Song	H01L 24/80
2022/0130761	A1 *	4/2022	Kim	H01L 23/5226
2022/0165721	A1 *	5/2022	Lee	H01L 24/08
2022/0216185	A1 *	7/2022	Kao	H01L 23/53228
2023/0260894	A1 *	8/2023	Hu	H10B 63/80 257/690

* cited by examiner

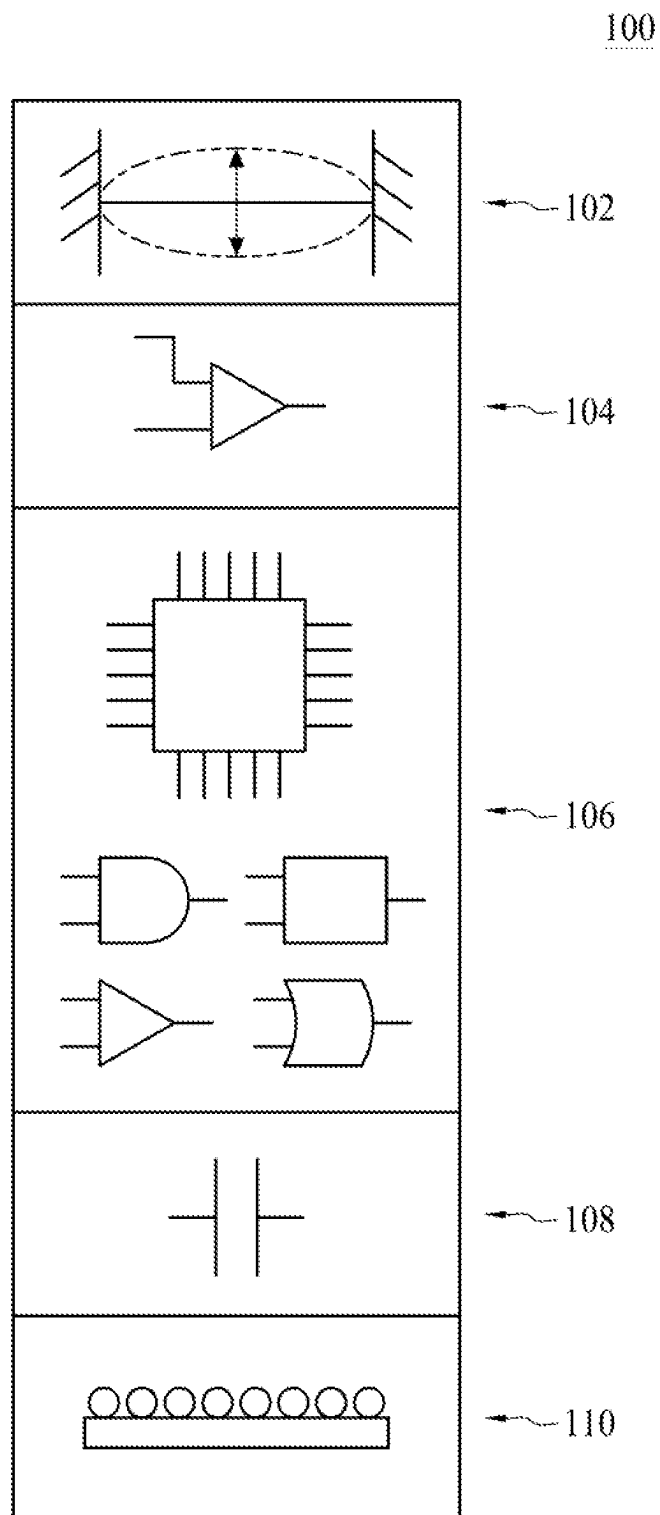


Figure 1

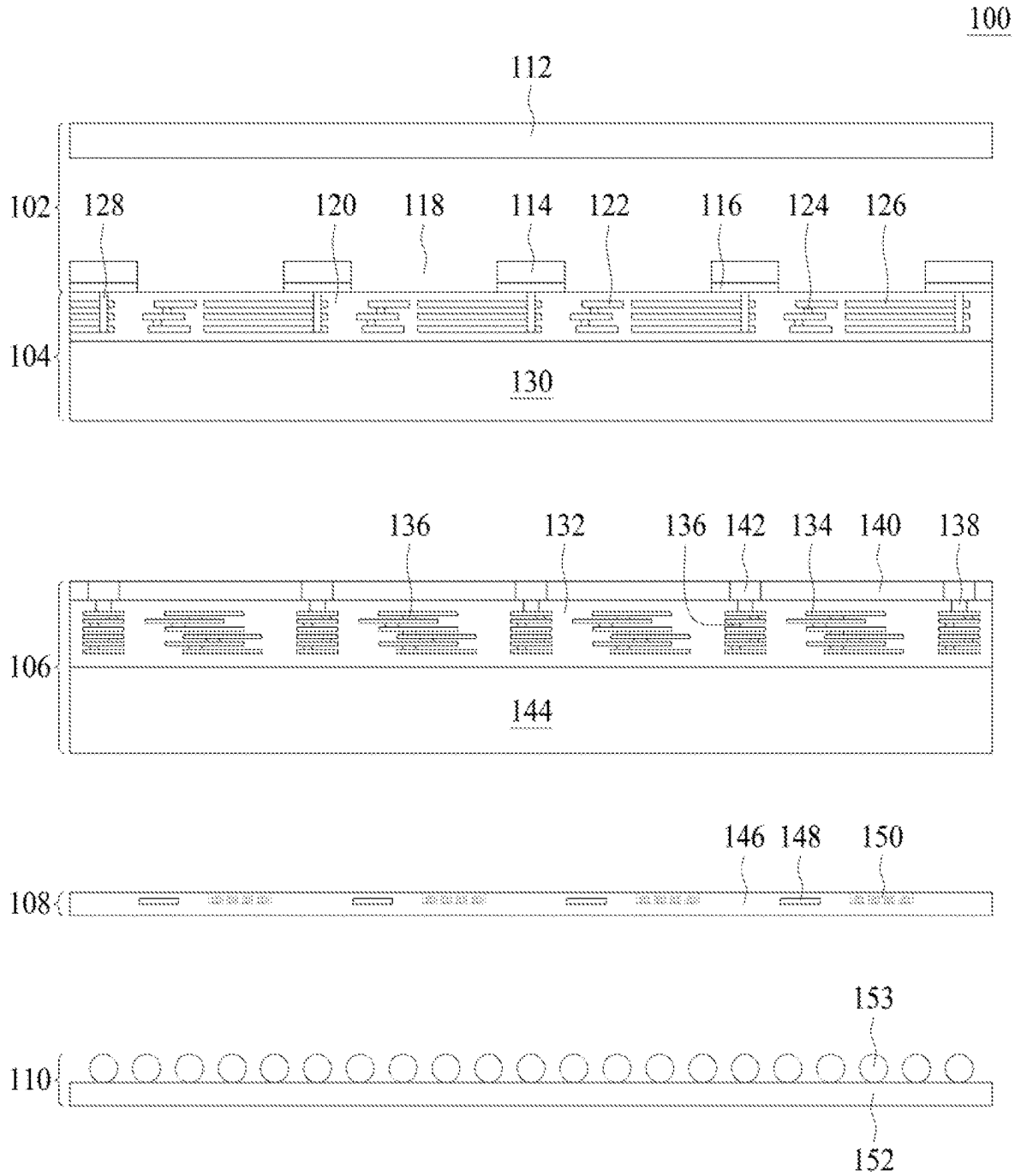


Figure 2A

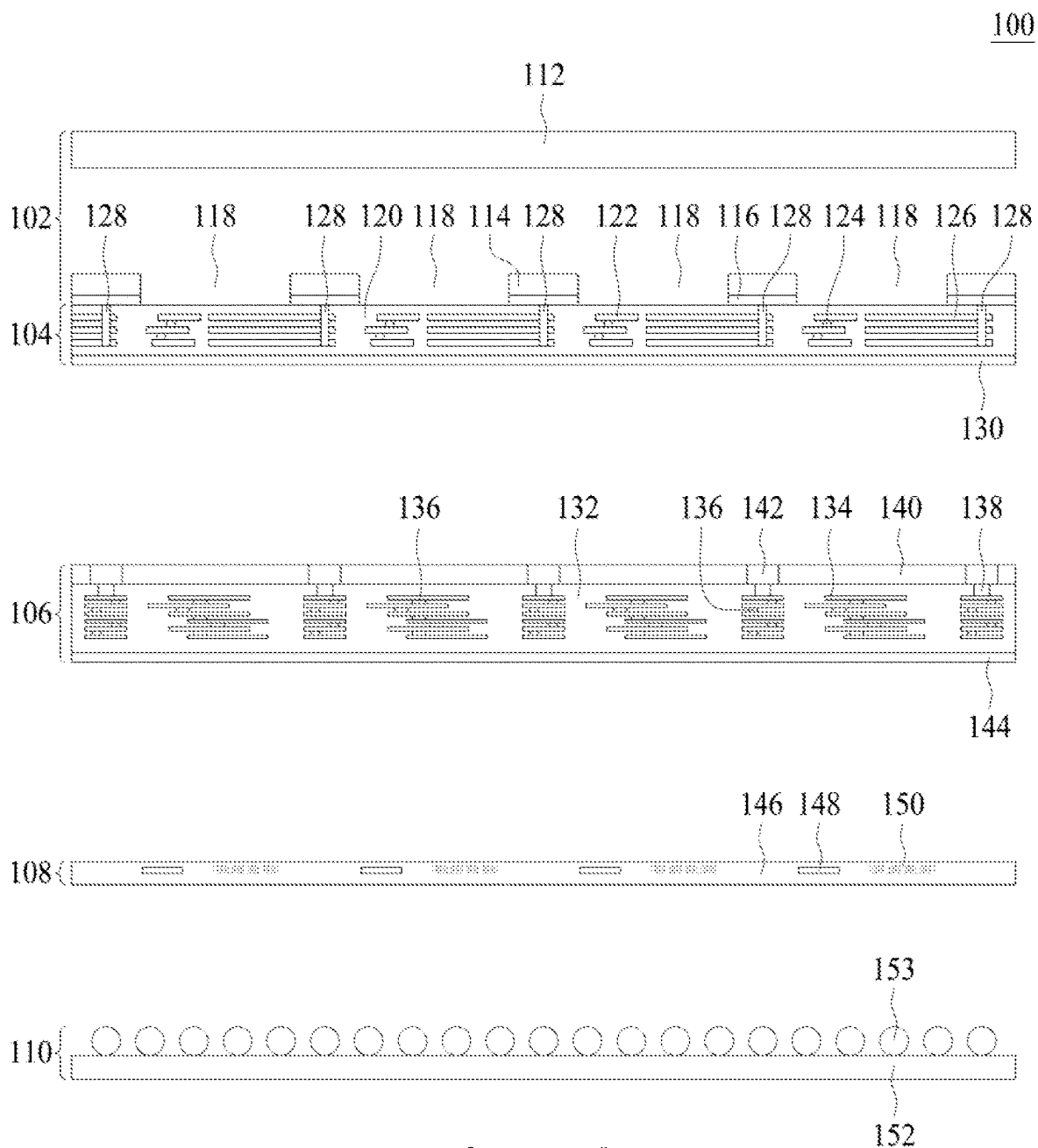


Figure 2B

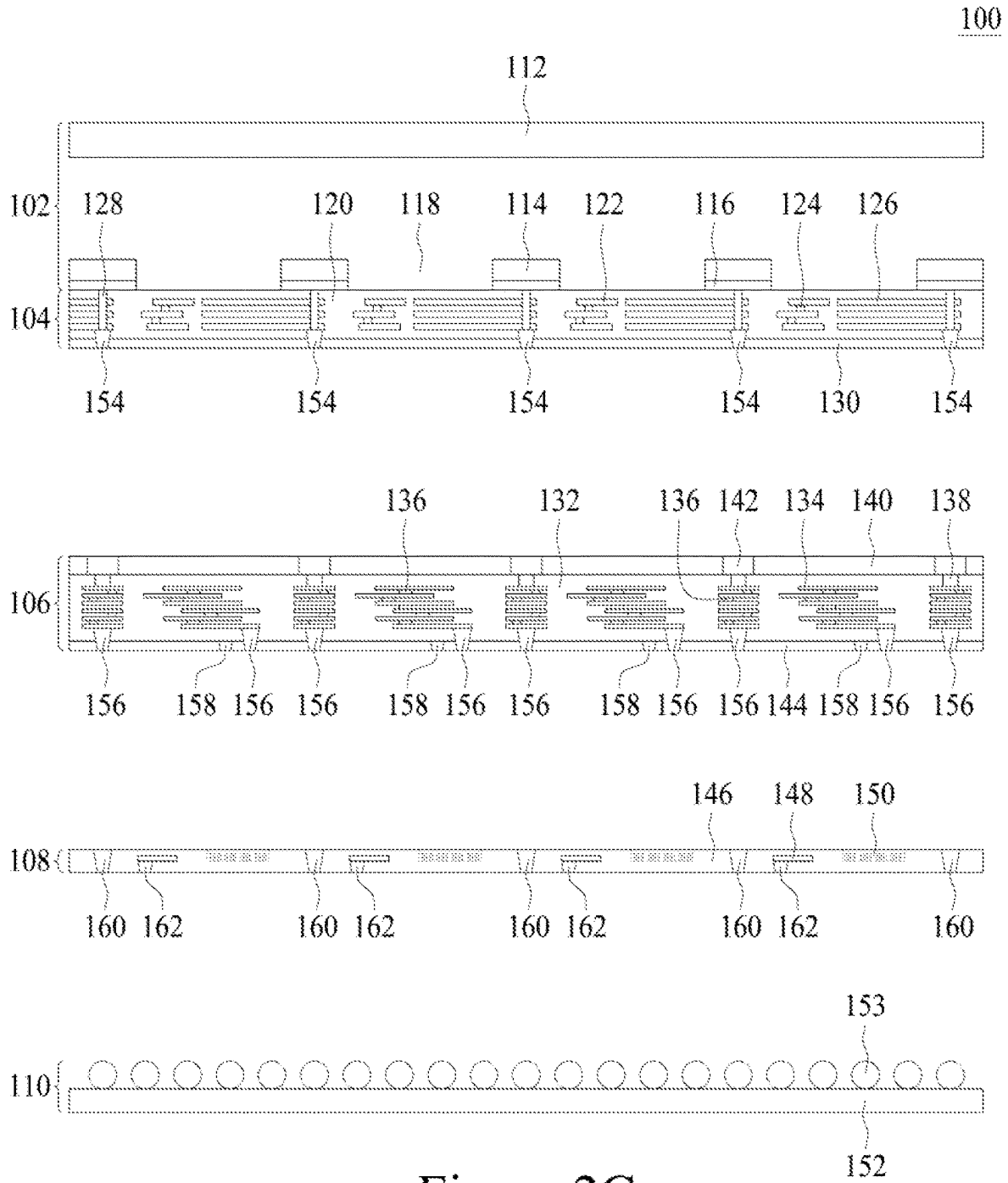


Figure 2C

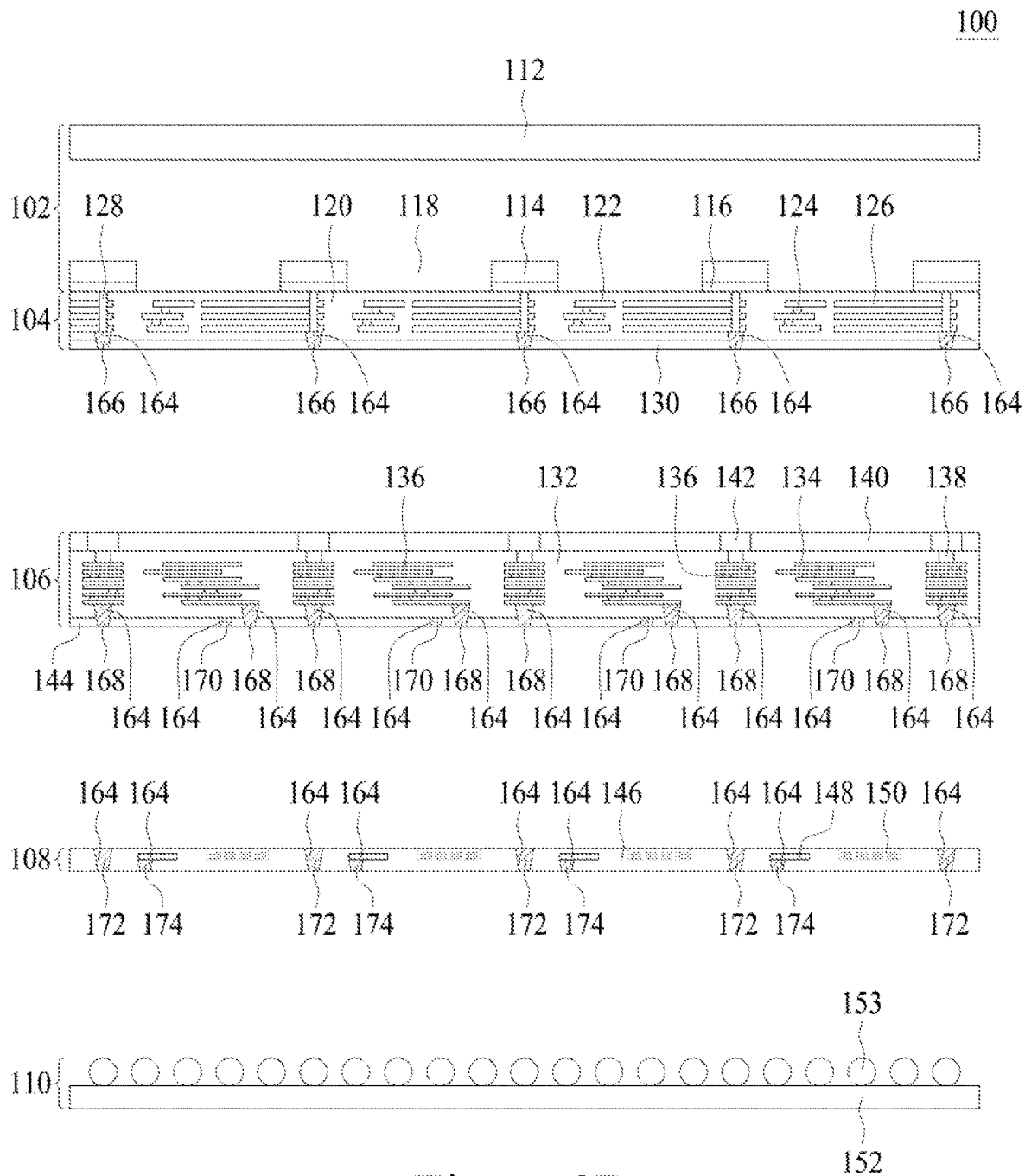


Figure 2D

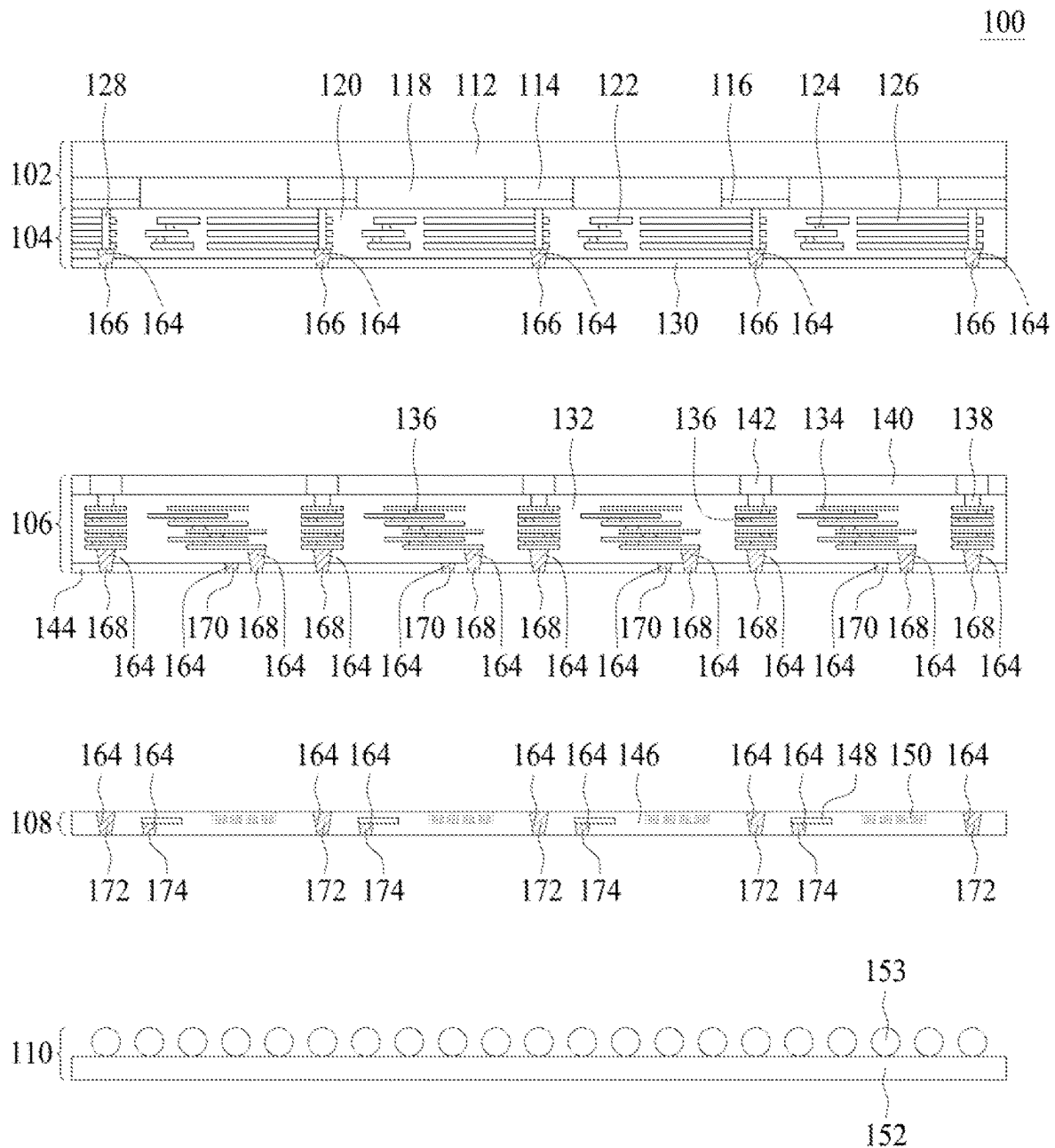


Figure 2E

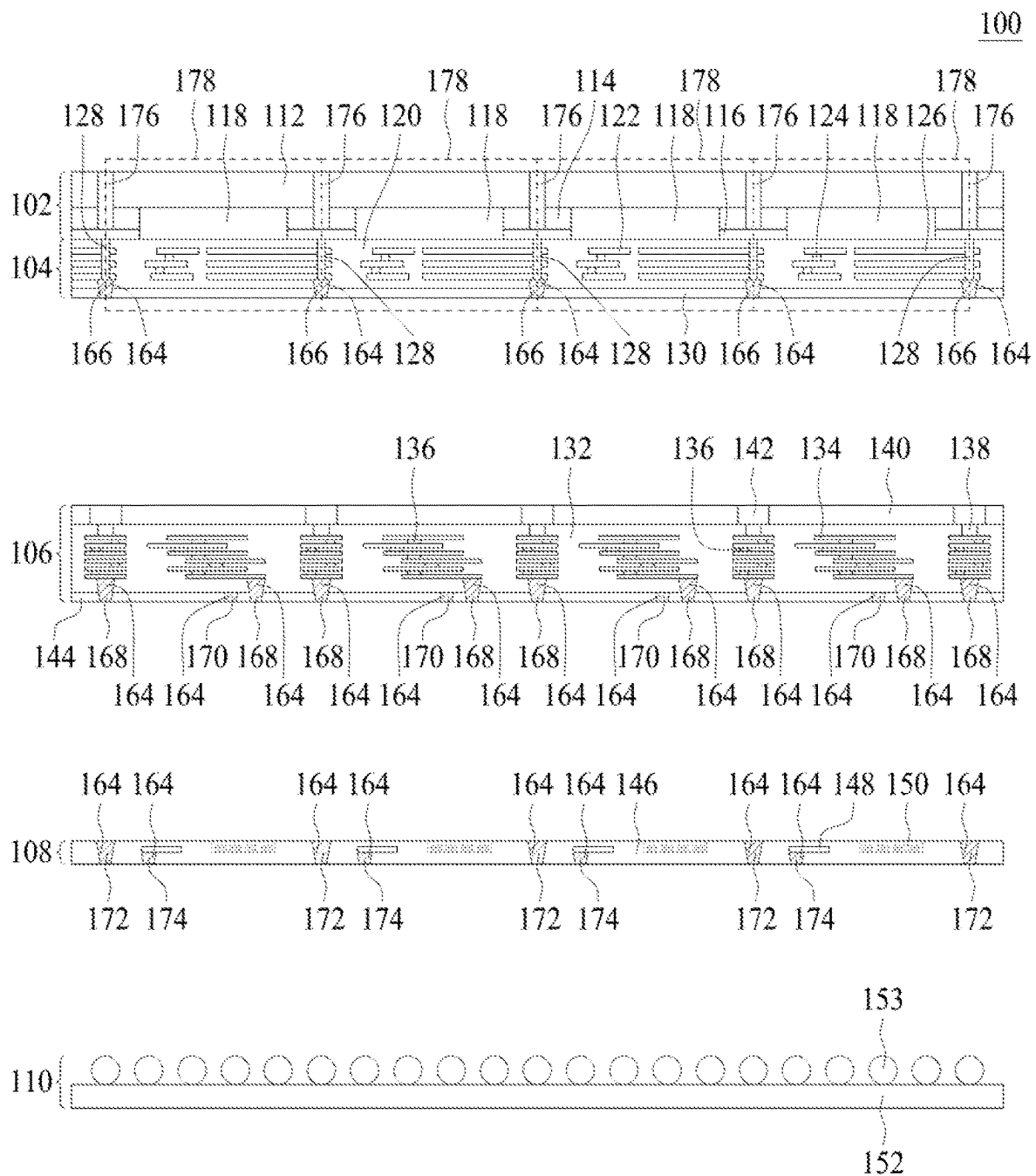


Figure 2F

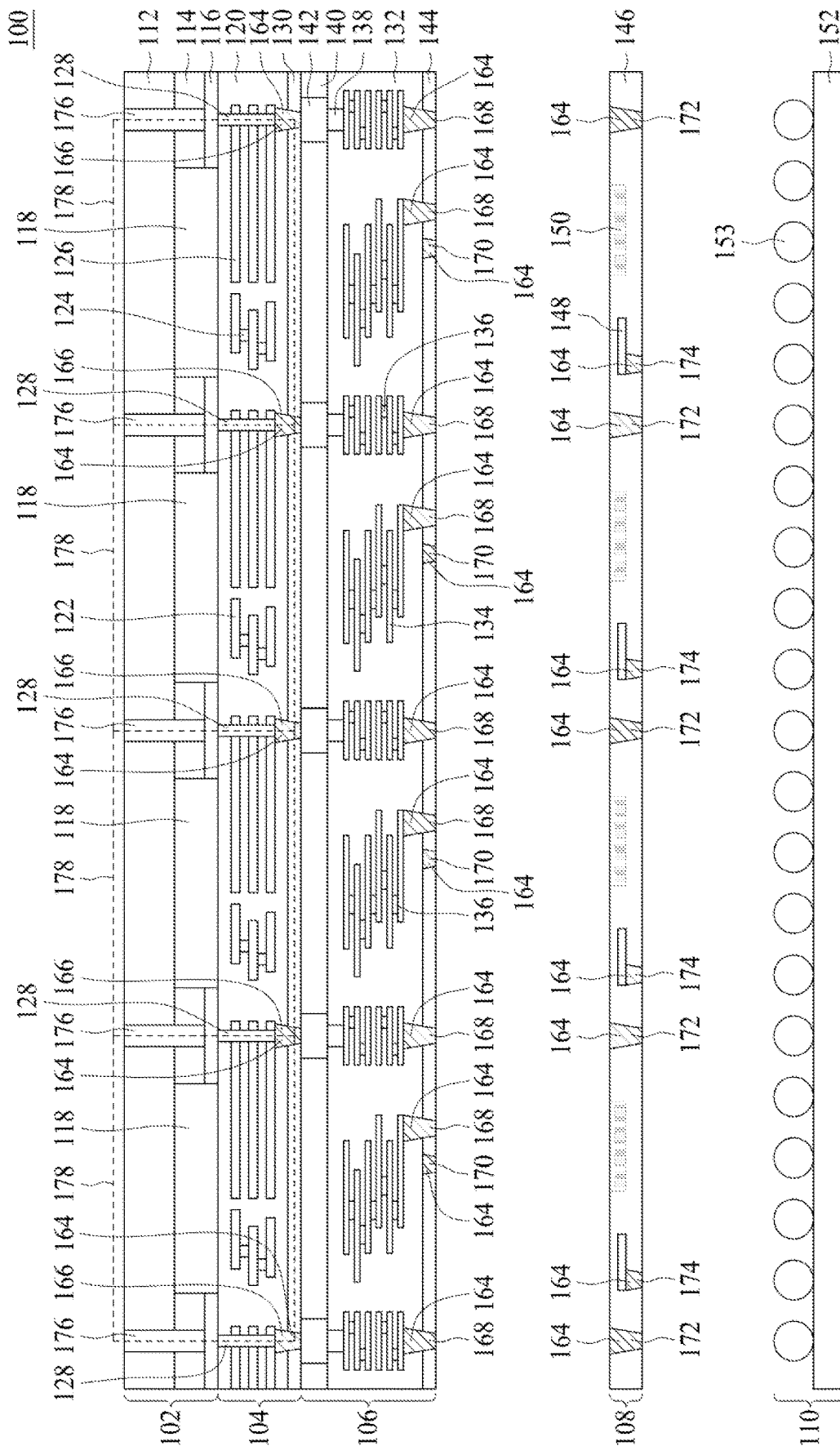


Figure 2G

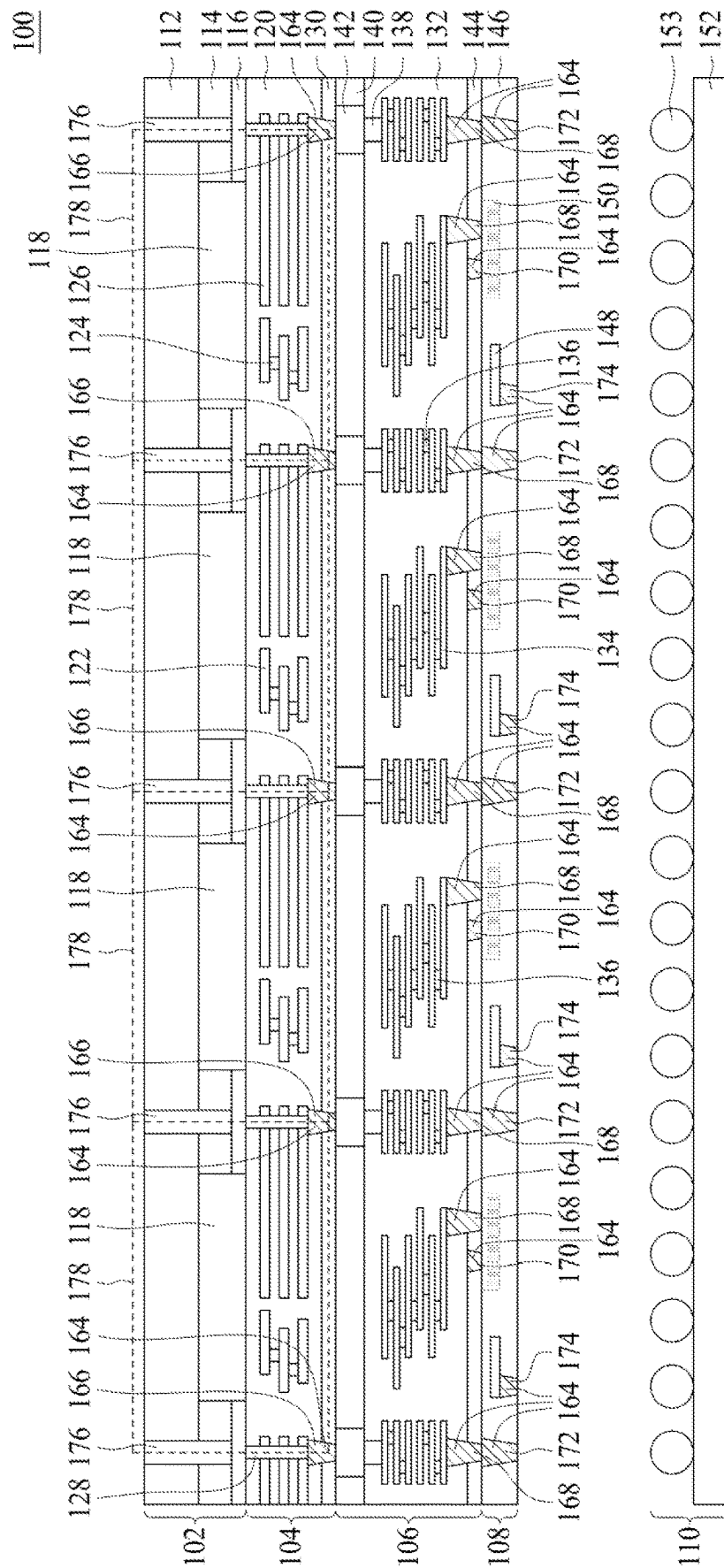


Figure 2H

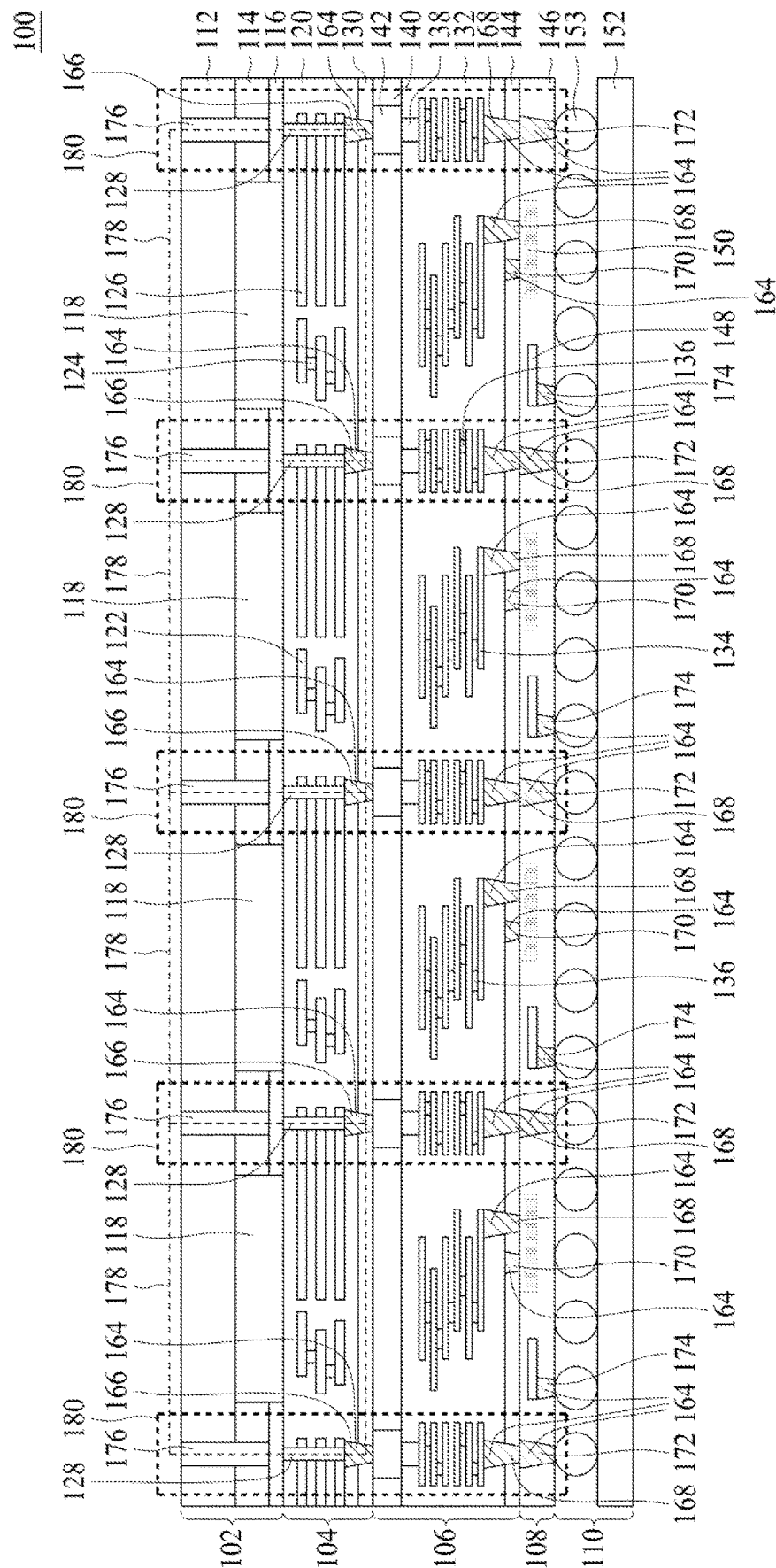


Figure 21

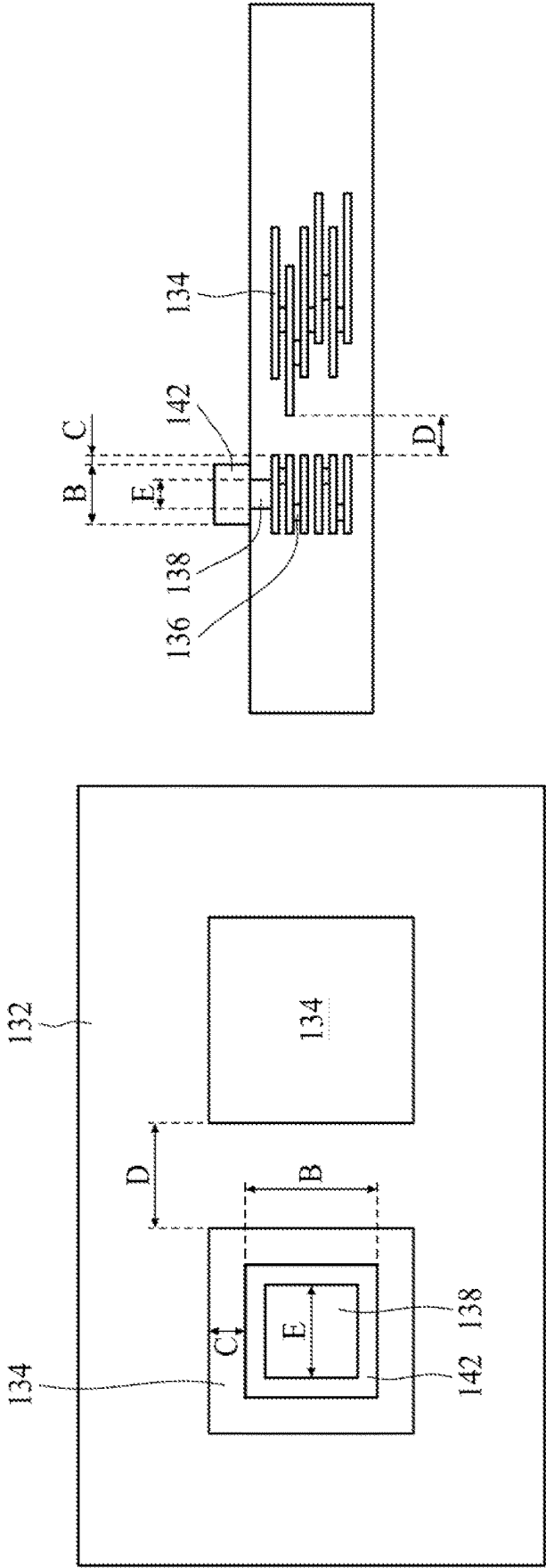


Figure 3

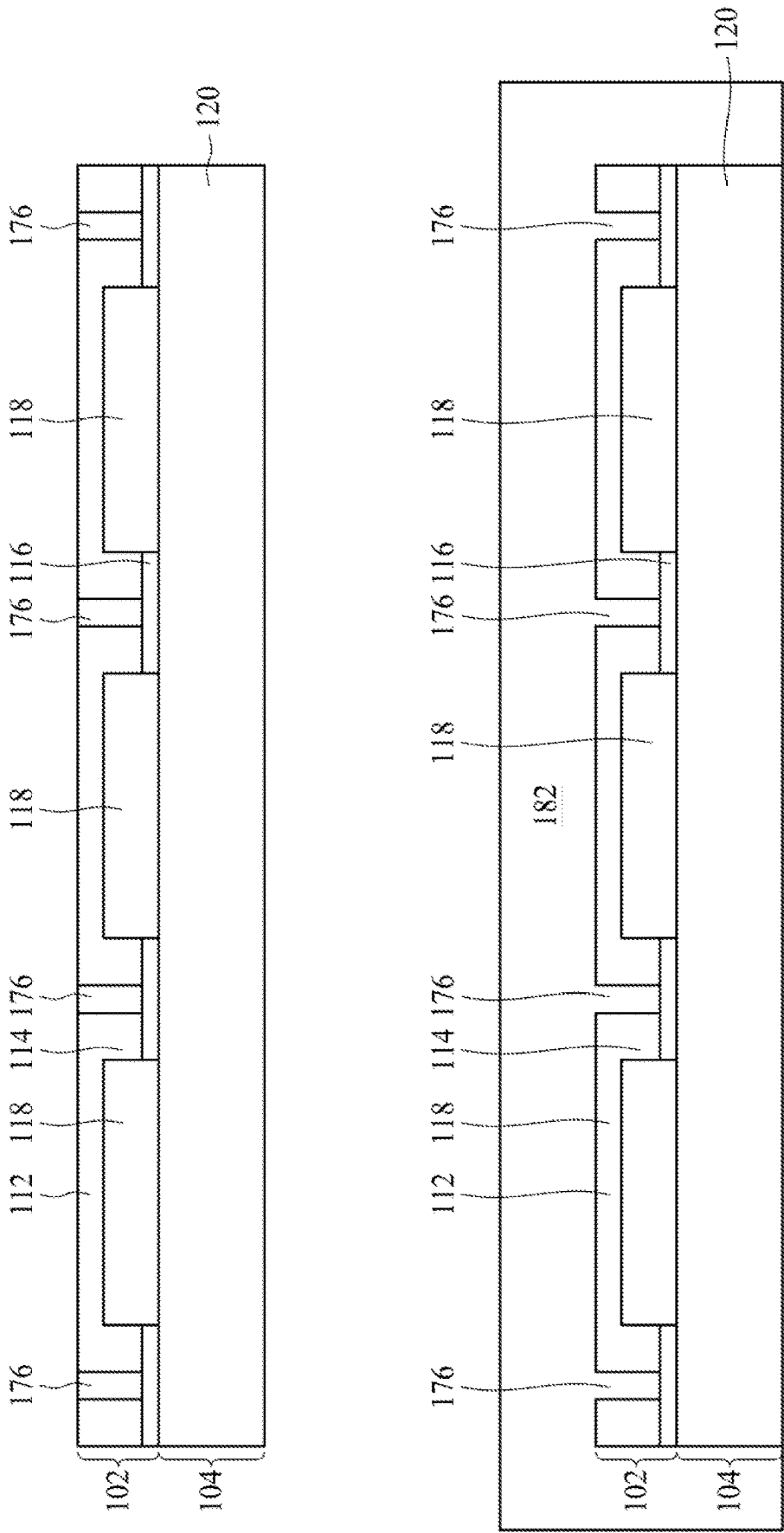


Figure 4

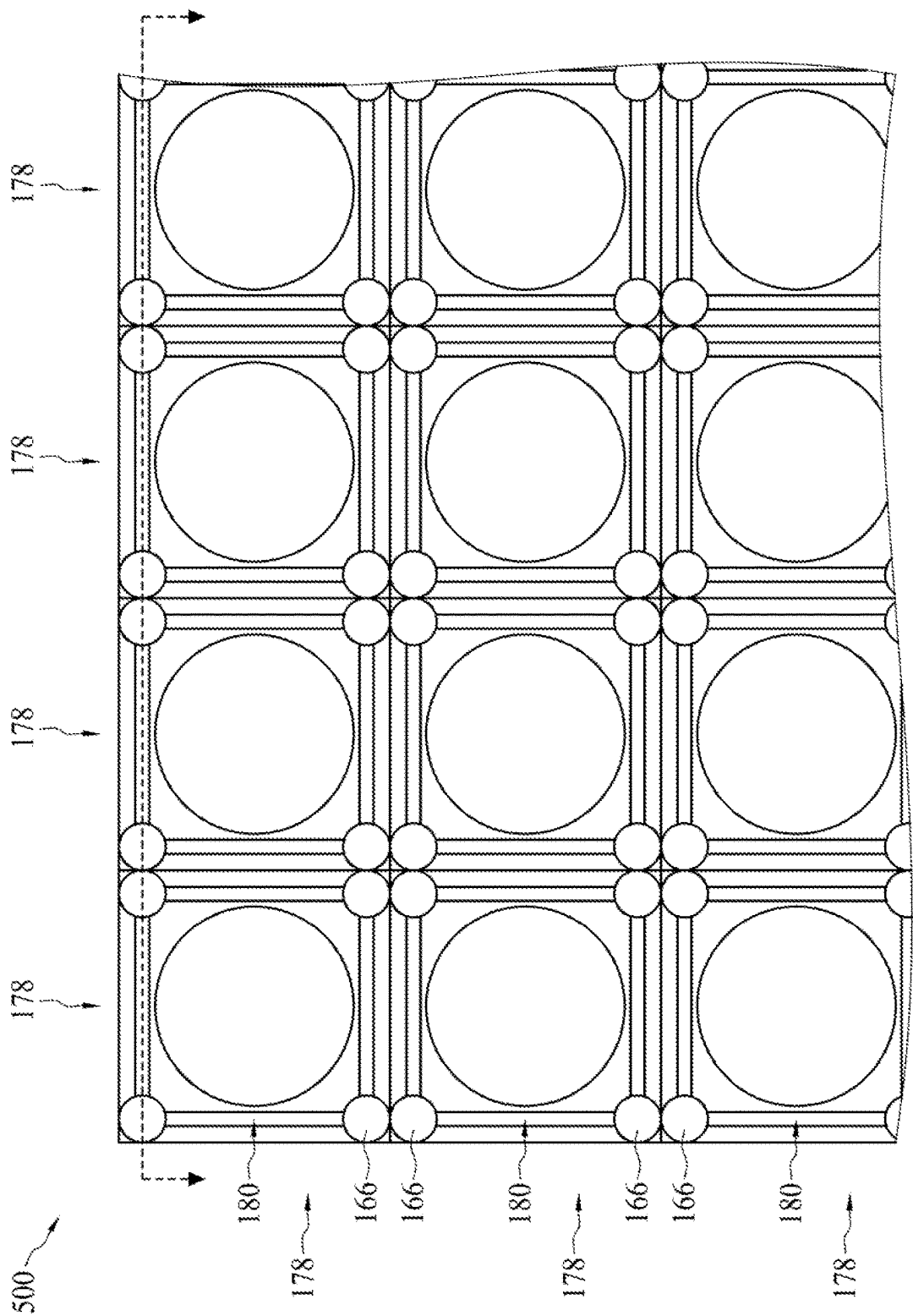


Figure 5A

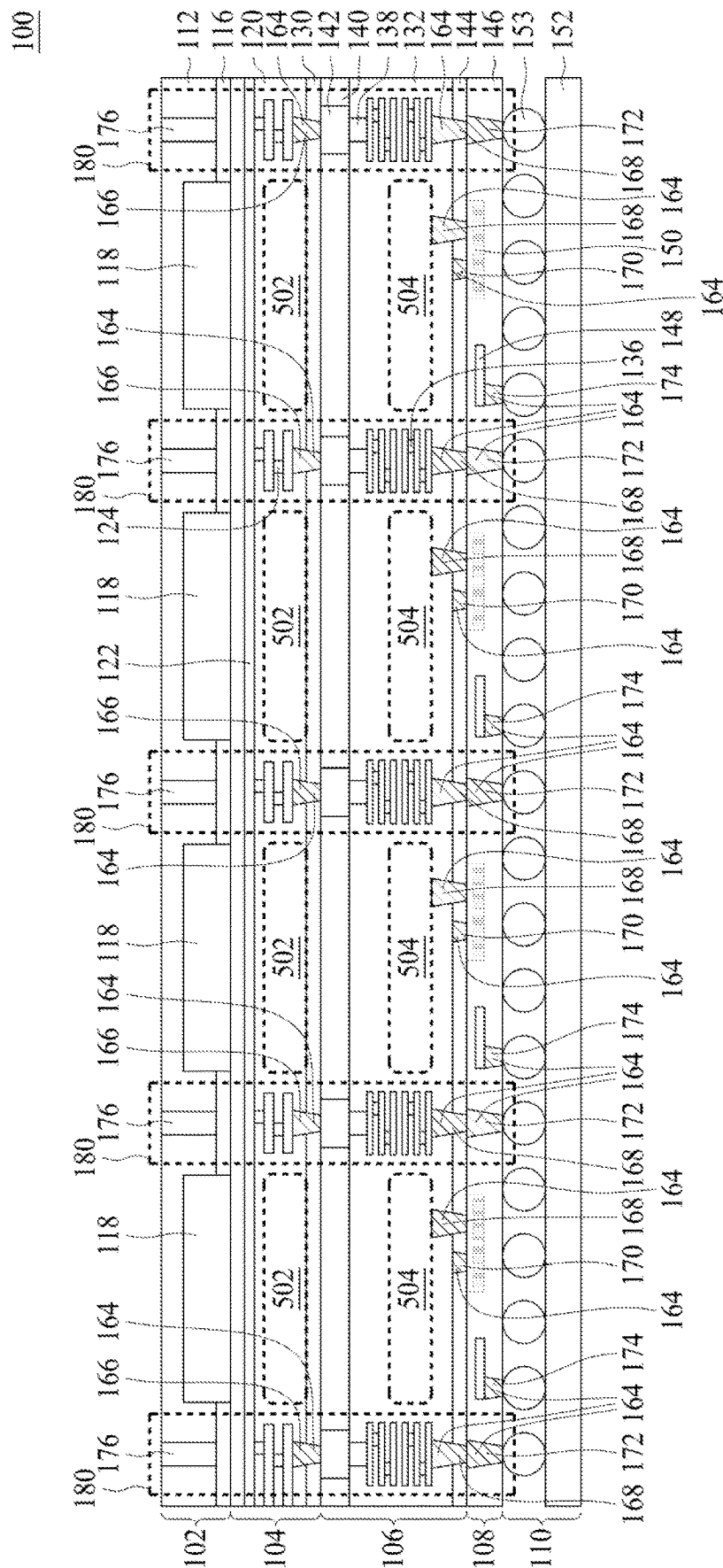


Figure 5B

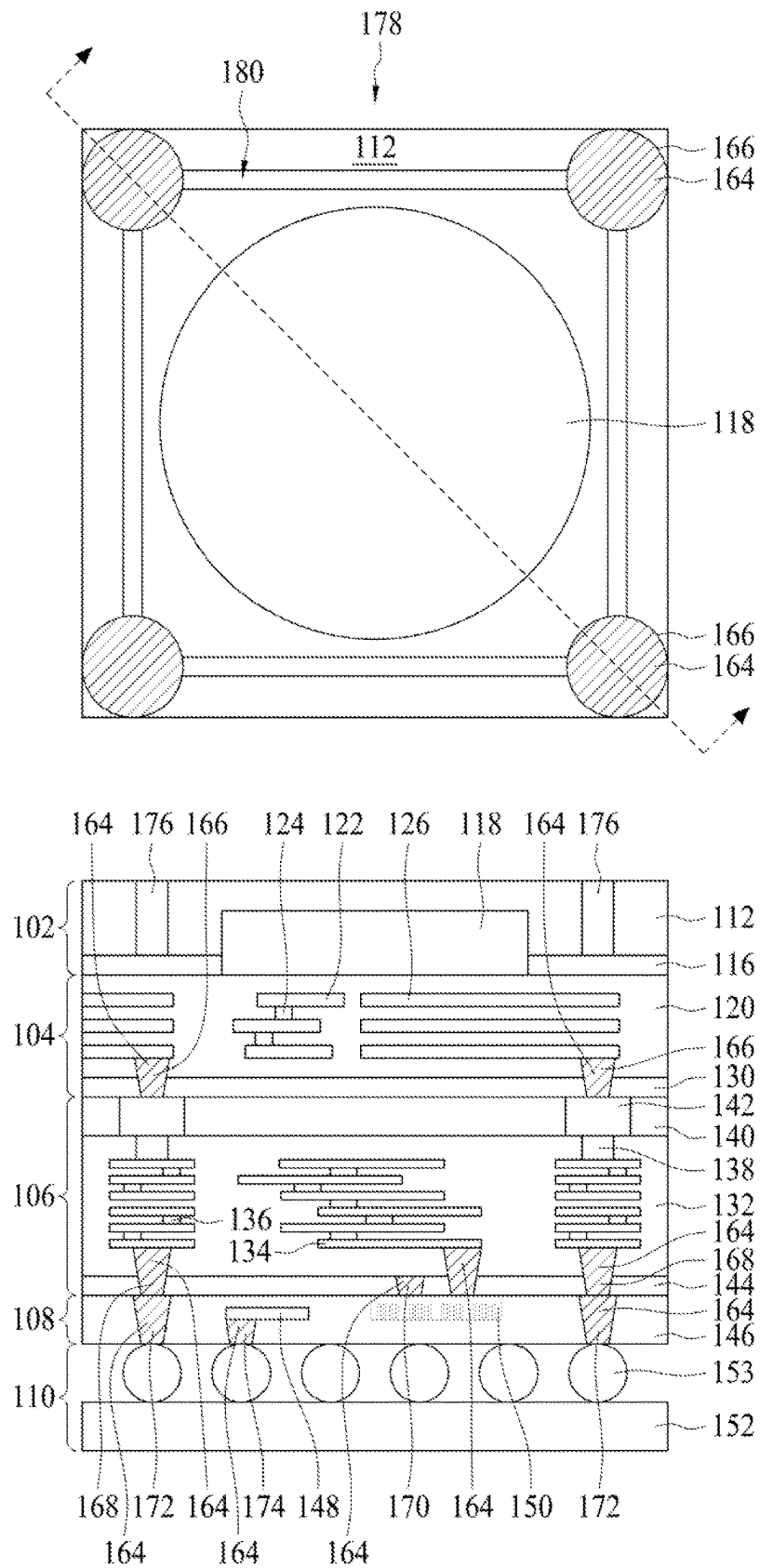


Figure 6A

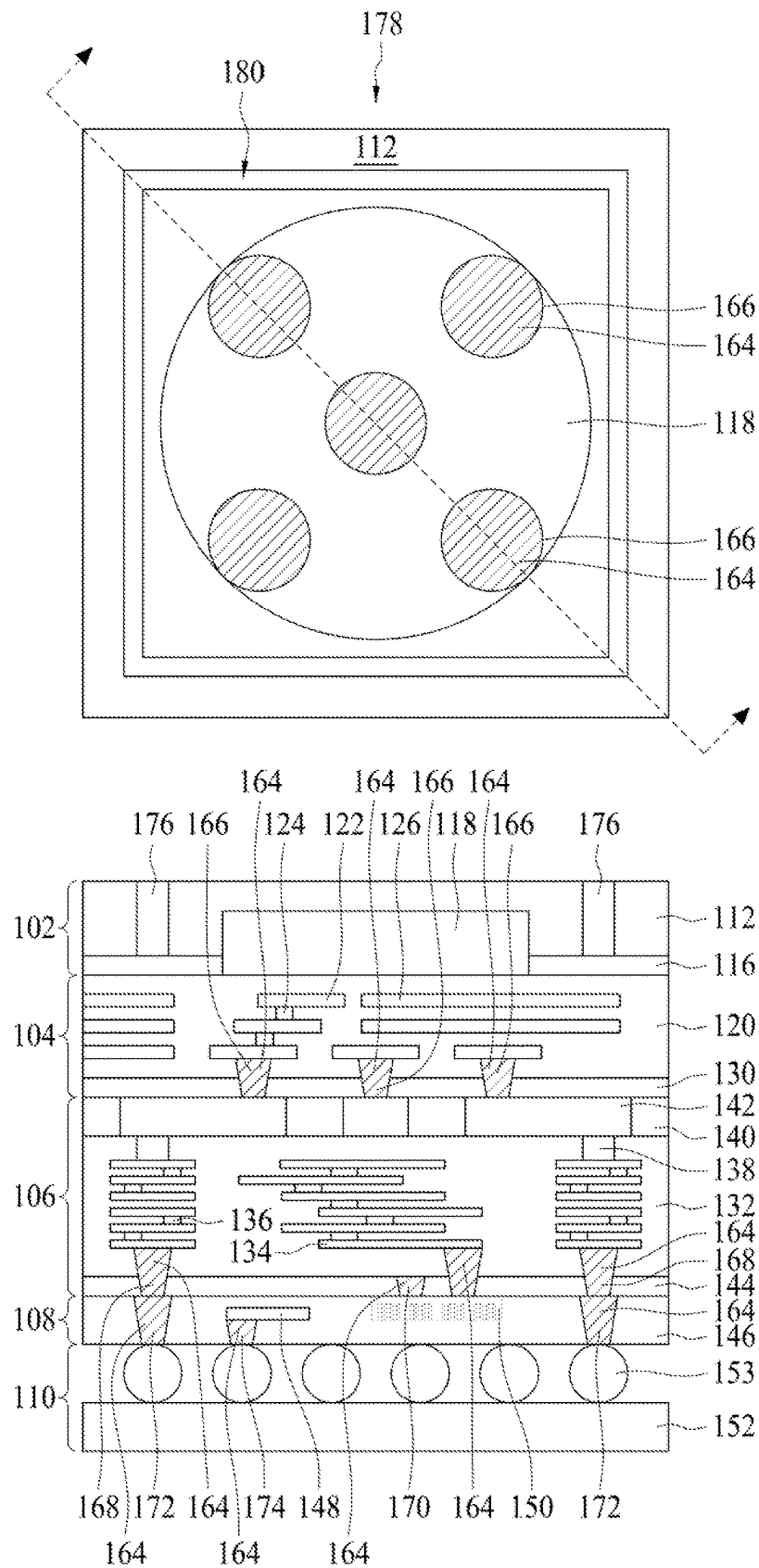


Figure 6B

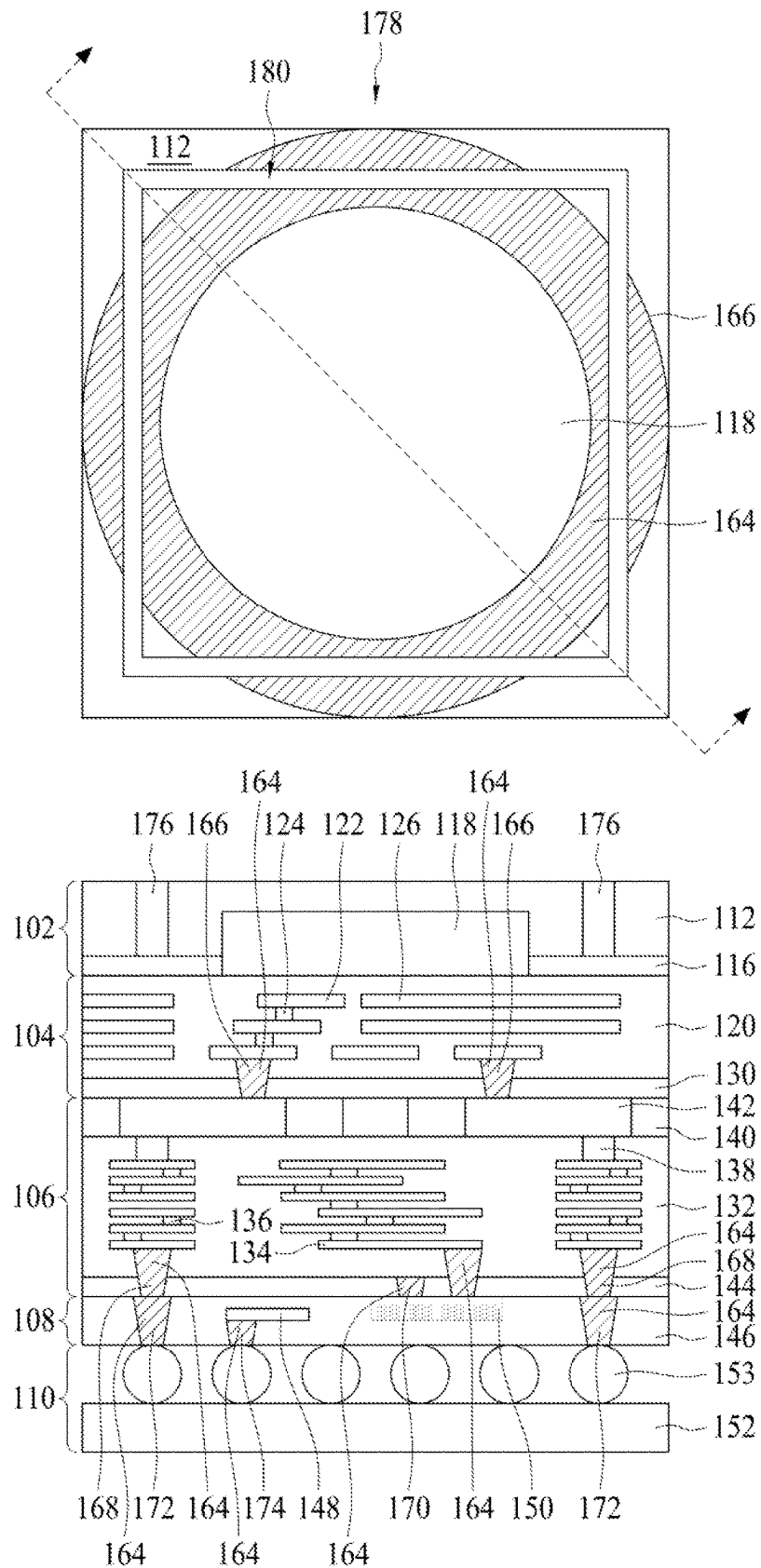


Figure 6C

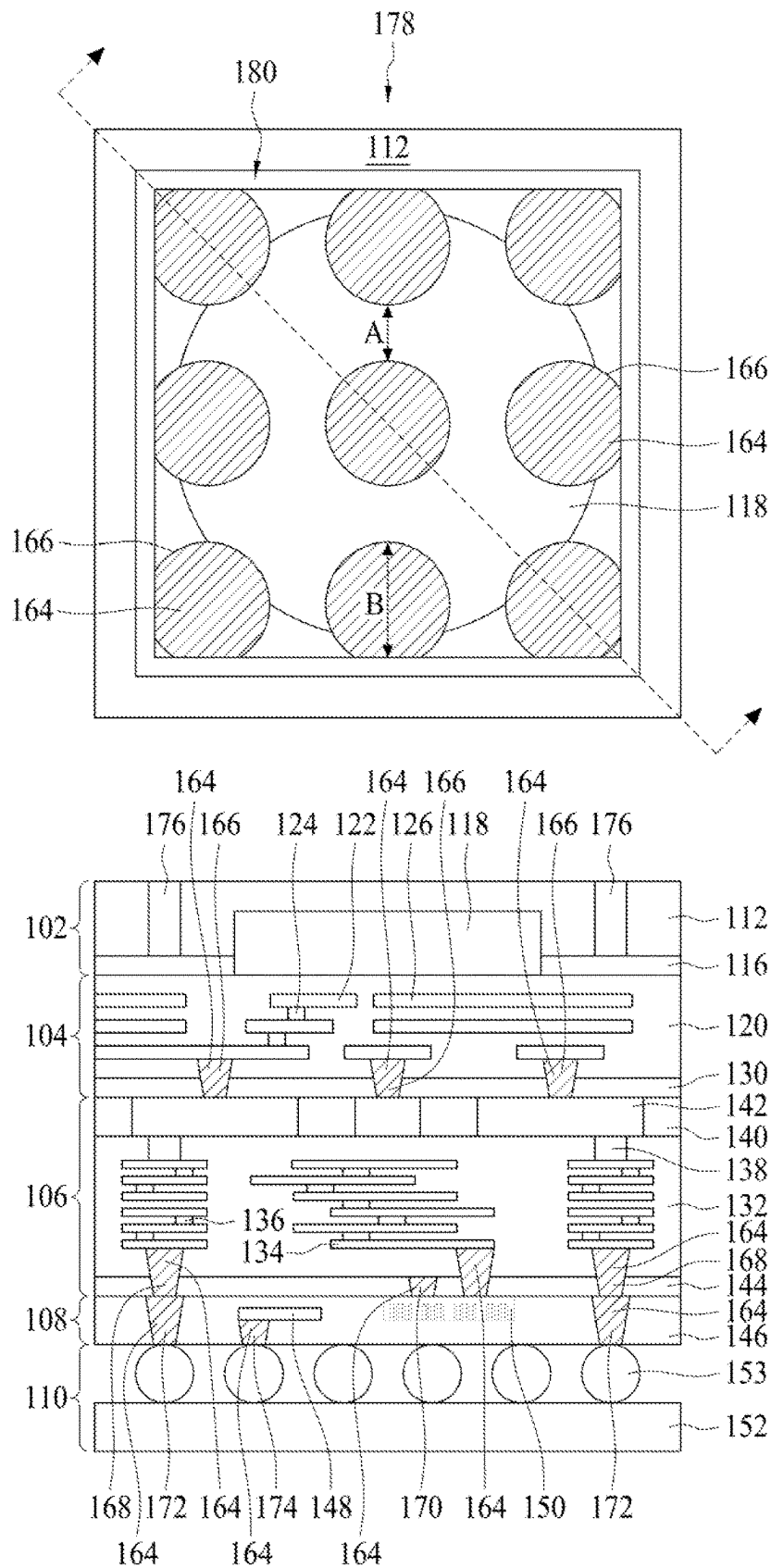


Figure 6D

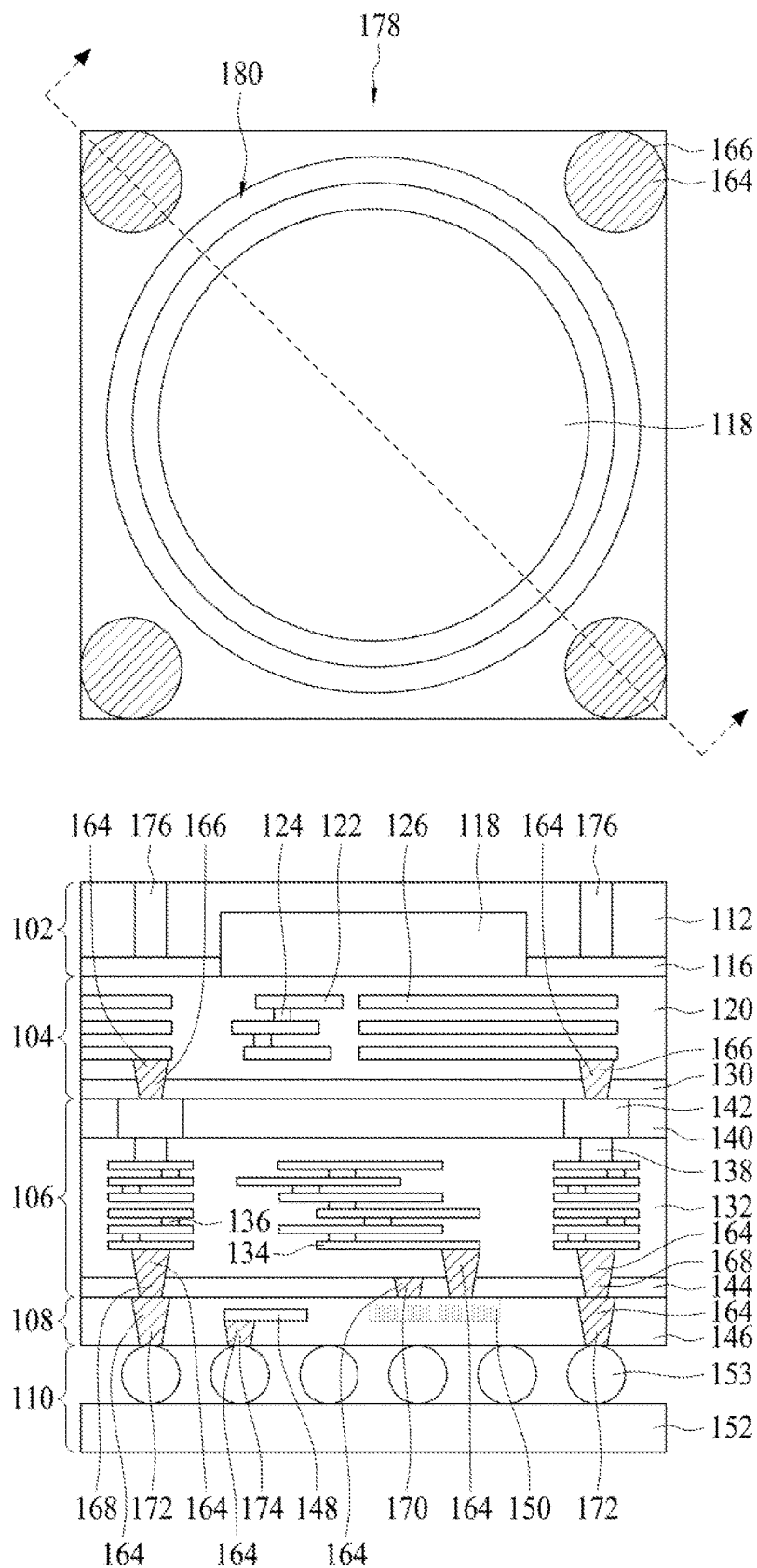


Figure 7A

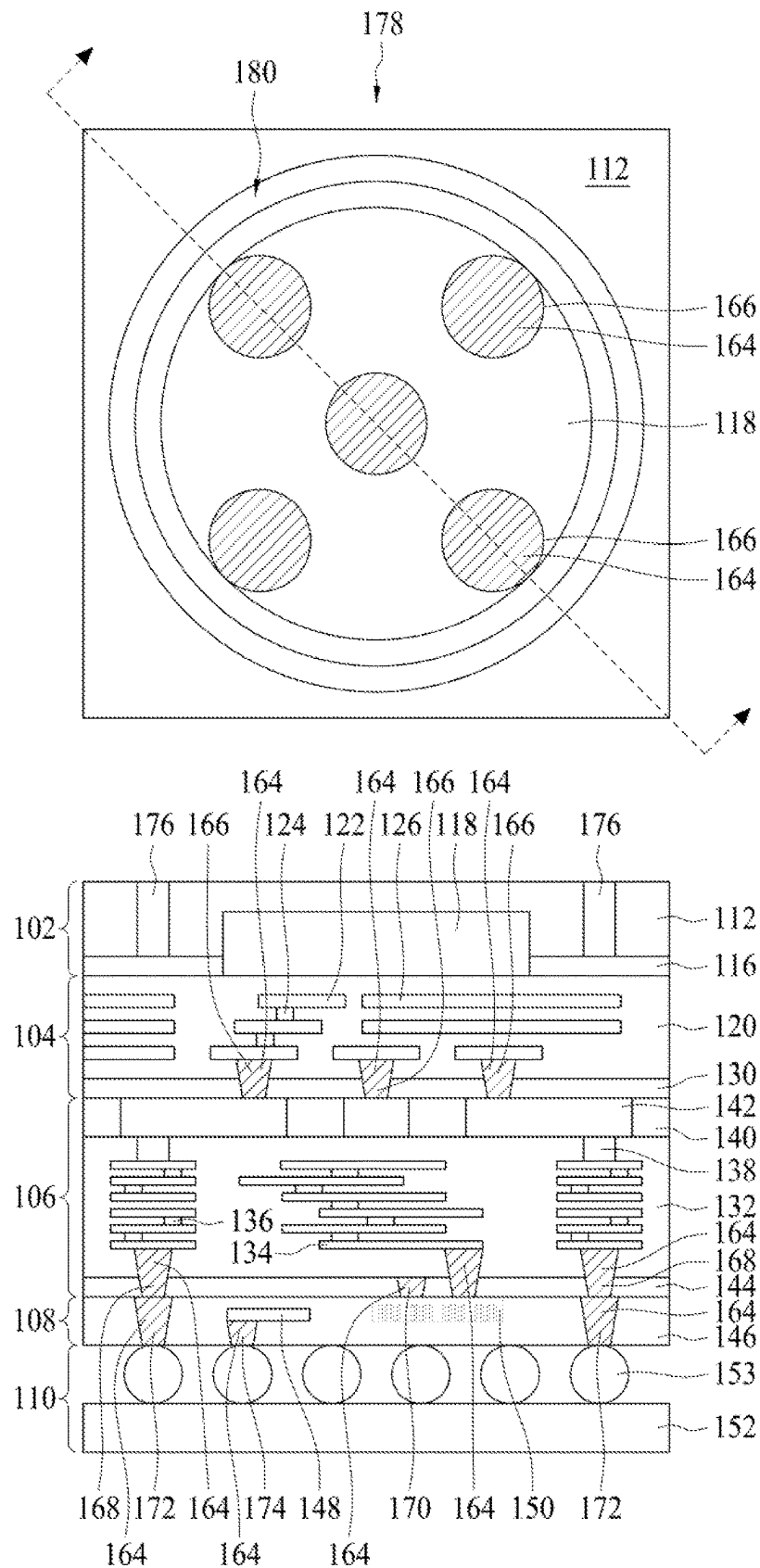


Figure 7B

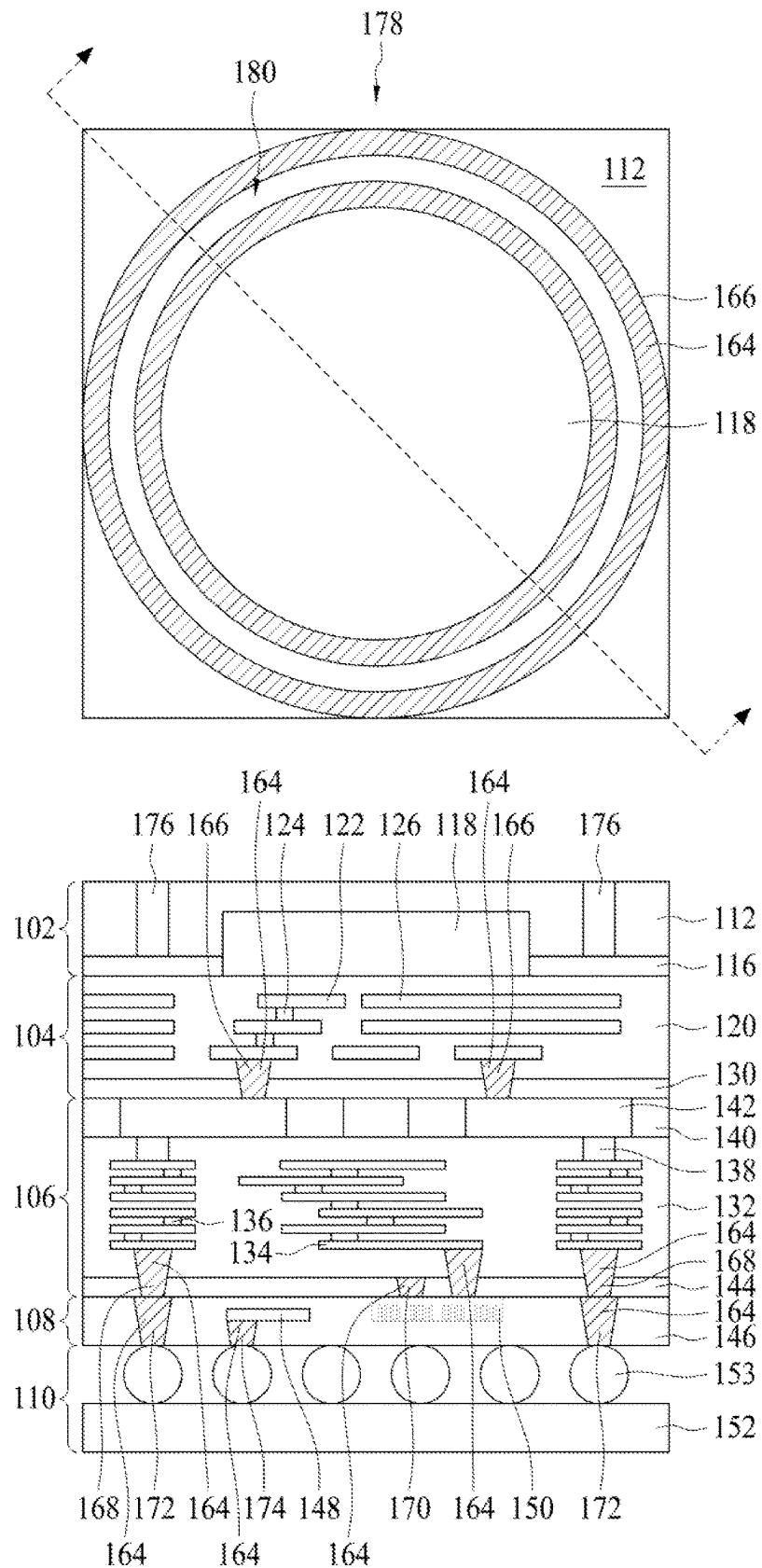


Figure 7C

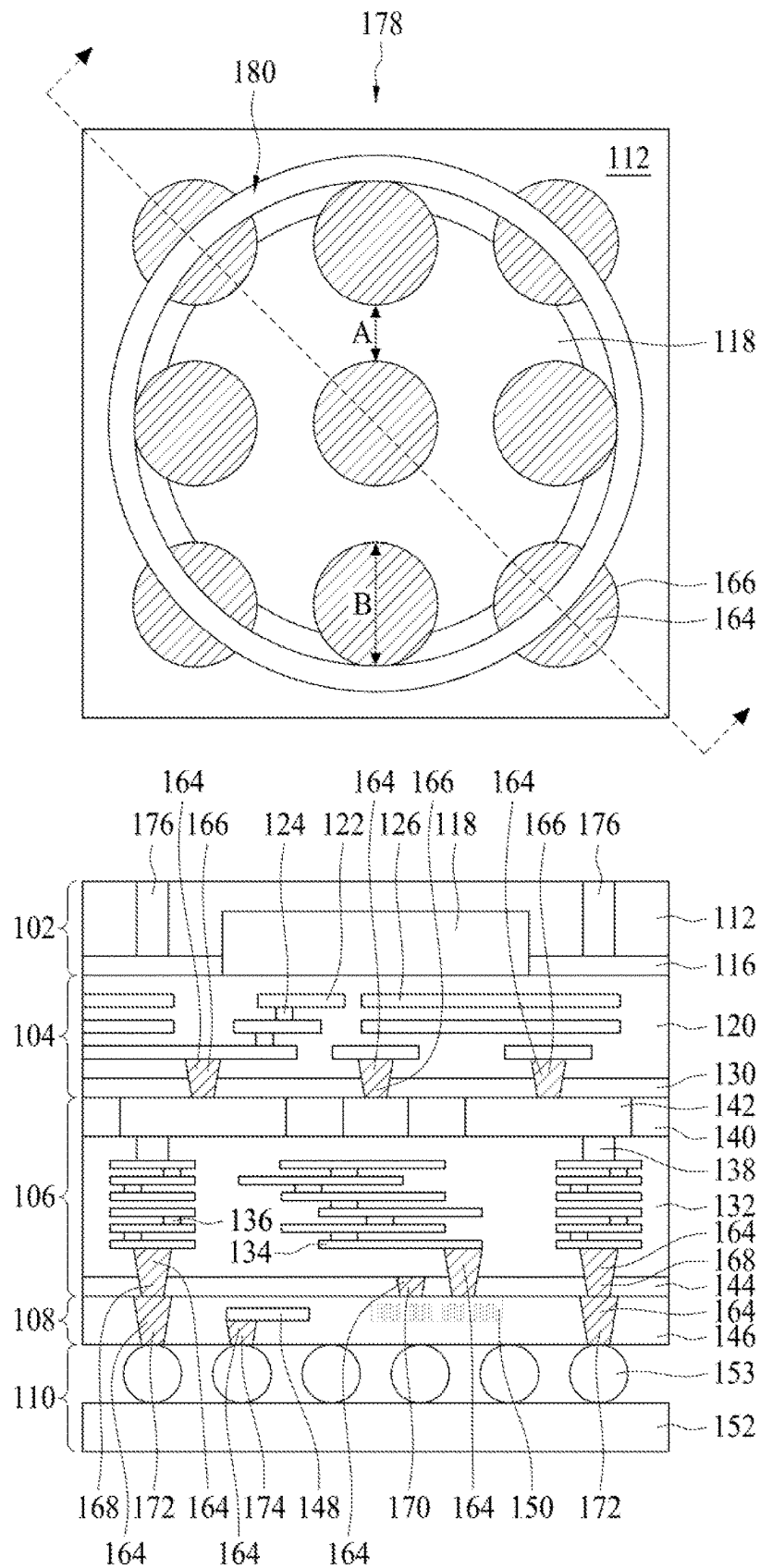


Figure 7D

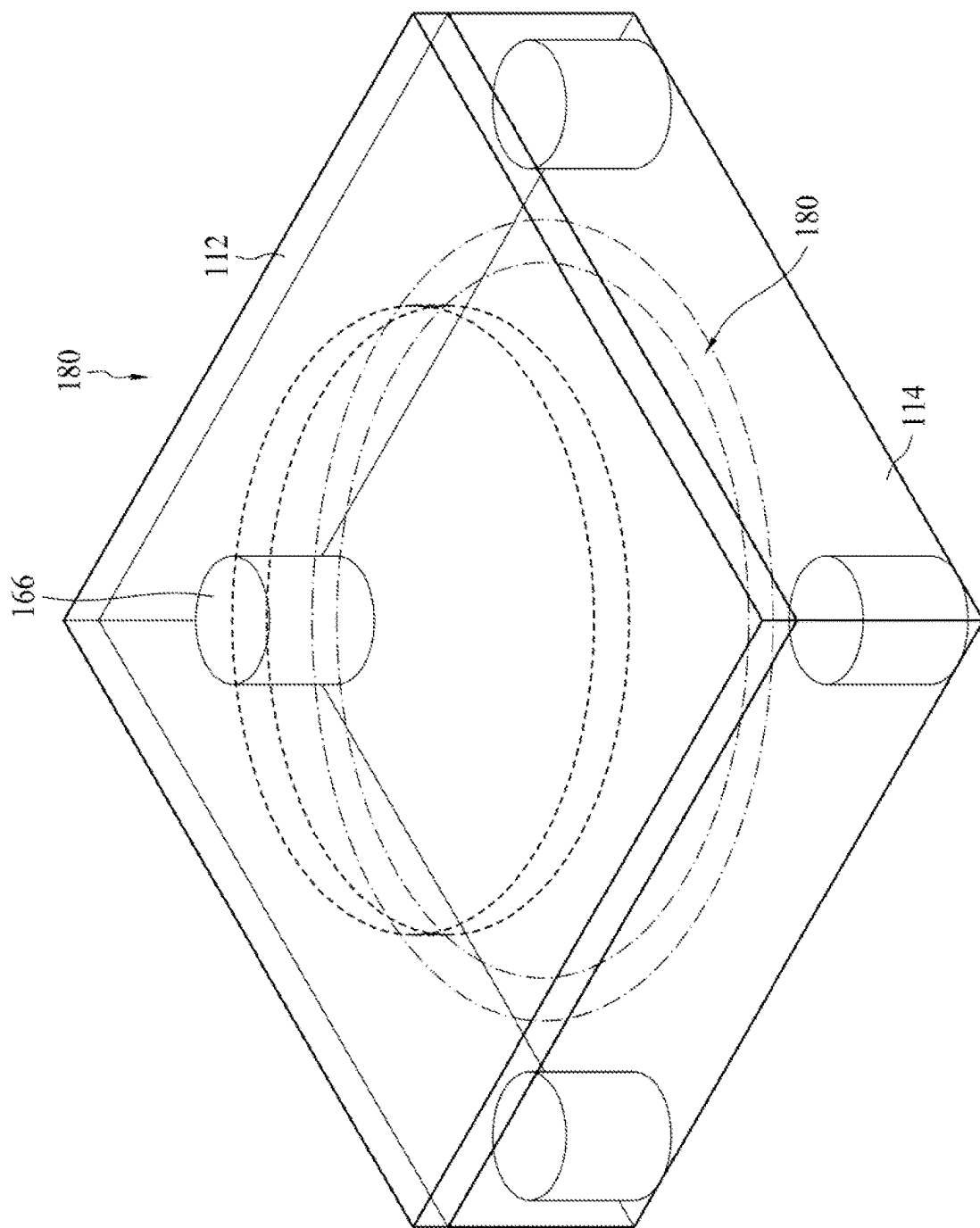


Figure 8

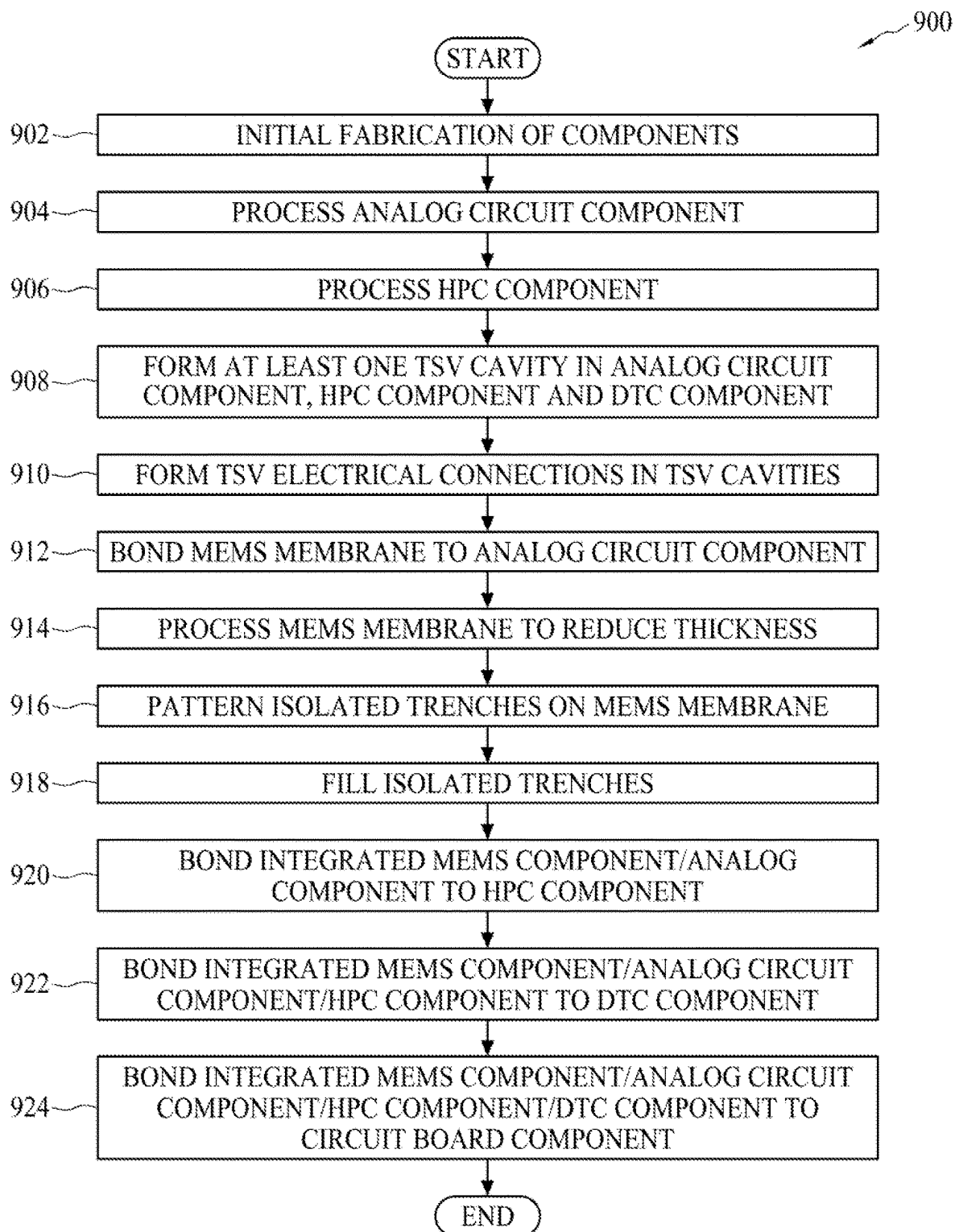


Figure 9

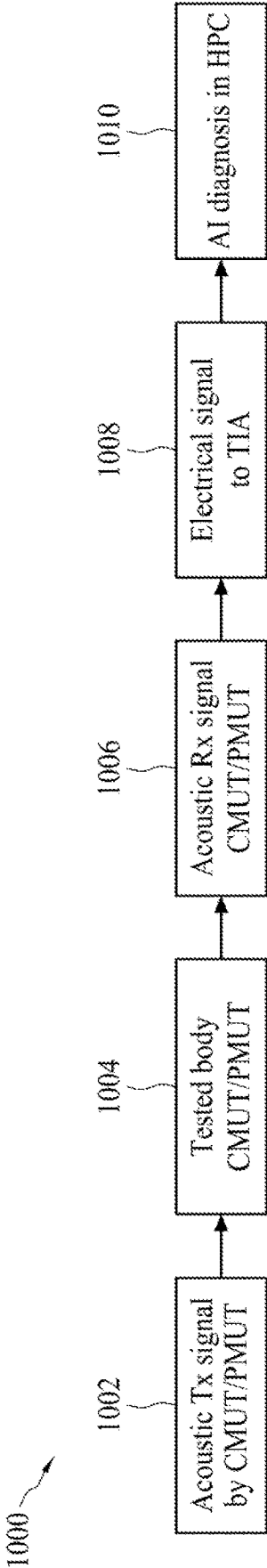


Figure 10

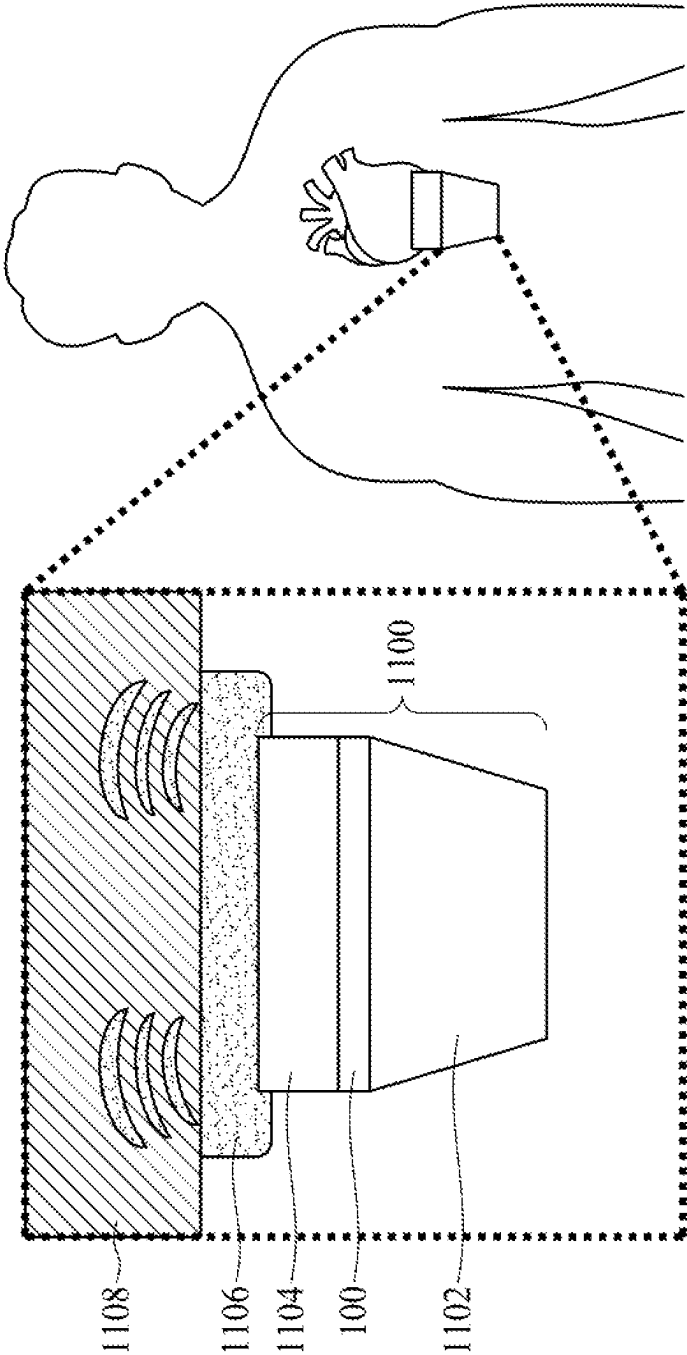


Figure 11

SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURE

BACKGROUND

Micro-electro-mechanical systems (MEMS) is a technology that utilizes miniature mechanical and electromechanical elements (e.g., devices or structures) on an integrated circuit substrate. MEMS devices may range from relatively simple structures with no moving elements, to complex electro-mechanical systems utilizing a variety of moving elements under the control of an integrated microelectronic controller. The devices or structures that are used in MEMS include microsensors, micro-actuators, microelectronics, and microstructures. MEMS devices may be used in a wide range of applications, including, for example and without limitation, motion sensors, pressure sensors, inertial sensors, micro-fluidic devices (e.g., valves, pumps, nozzle controls), optical devices, imaging devices (e.g., micromachined ultrasonic transducers (MUT's)), capacitive ultrasound transducers (CMUT), and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a block diagram of a semiconductor device in accordance with some embodiments.

FIGS. 2A-2I are cross-sectional views of stages of manufacturing a semiconductor device in accordance with some embodiments.

FIG. 3 is a top view and side view of a bond pad, bond pad via, and HPC metal components in accordance with some embodiments.

FIG. 4 is a comparative view of embodiments of an isolated trench in accordance with some embodiments.

FIG. 5A is an illustrative top view of MEMS array in accordance with some embodiments.

FIG. 5B is a simplified cross-sectional view of the MEMS array of FIG. 5A.

FIGS. 6A-6D are top and cross-sectional views of MEMS pixel patterns in accordance with some embodiments.

FIGS. 7A-7D are top and cross-sectional views of MEMS pixel patterns in accordance with some embodiments.

FIG. 8 is an isometric view of a MEMS pixel in accordance with some embodiments.

FIG. 9 is a method for forming a semiconductor device in accordance with some embodiments.

FIG. 10 is an example operation of a semiconductor device in accordance with some embodiments.

FIG. 11 is an example implementation of the semiconductor device of FIG. 10.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature

in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The term "about" can be used to include any numerical value that can vary without changing the basic function of that value. When used with a range, "about" also discloses the range defined by the absolute values of the two endpoints, e.g., "about 2 to about 4" also discloses the range "from 2 to 4." The term "about" may refer to plus or minus 10% of the indicated number.

In some embodiments, a semiconductor device (e.g., integrated chip) comprises a microelectromechanical systems (MEMS) device. The MEMS device comprises a cavity and a movable membrane. The configuration (e.g., structural configuration) of the MEMS device is dependent on the type of MEMS device. The present disclosure discusses structures and fabrication methods of a capacitive micromachined ultrasonic transducer (CMUT). The CMUT device is widely used in high-resolution applications, e.g., medical diagnostics, imaging, sensors, etc., as well as air-coupled non-destructive evaluation, ultrasonic flow meters for narrow gas pipelines, microphones with RF detection, Lamb wave devices, smart microfluidic channels, and the like. Current production of CMUTs utilizes a bottom electrode covered by a dielectric film, with the movable membrane spaced apart from the bottom electrode by a gap (thus, the cavity of the CMUT is located between the movable membrane and the bottom electrode). The movable membrane carries a top electrode, so that movement of the membrane in response to sonic waves creates a variable capacitance between the bottom and top electrodes, thus providing an acoustic transducer. (Conversely, an AC electrical signal applied across the top and bottom electrodes can cause the membrane to oscillate and generate an acoustic wave).

Use of such MEMS devices in ultrasound equipment involves an array of CMUTs controlled by a central processing unit, generally in communication with the CMUT array via a communications cable that supplies control signals, power, and receives output from the array. The central processing unit then processes the returned signals to output a visual representation of an object on an associated display, e.g., inside of a food container, pregnancy ultrasounds, luggage contents, etc. These kiosk-like pieces of equipment require remote processing capabilities and are generally bulky, requiring frames and structures to allow movement of the system. Further, the costs associated with this system include both the central processing unit (generally implemented as a desktop, laptop, or the like, computing

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system), the frame, the display, and the ultrasound wand (i.e., the wired component that contains the CMUT array), may be substantial.

Turning now to FIG. 1, there is shown a representative block diagram of a semiconductor device **100** in accordance with one embodiment of the subject application. As shown in FIG. 1, the semiconductor device **100** includes a MEMS component **102**, an analog circuit component **104**, a high performance chip (HPC) component **106**, a deep trench capacitor (DTC) component **108**, and a printed circuit board component **110**. The three-dimensional vertical integration of the semiconductor device **100** is illustrated in FIG. 1 for exemplary purposes and is intended solely to provide an overview of the semiconductor device **100** in accordance with one embodiment of the subject application.

In some embodiments, the MEMS component **102** may comprise, for example and without limitation, CMUT devices, PMUT devices, fluidic devices, accelerometers, or the like. The analog circuit component **104** may comprise various analog signal devices, analog electronic devices, or the like. Such example components of the analog circuit component **104** include, for example and without limitation, high voltage (HV) pulsers, transimpedance amplifiers (TIA), metal insulator metal (MIM), and the like. The HPC component **106** may be implemented with a variety of devices for processing, including, for example and without limitation, advanced transistors (e.g., FinFET, gate all-around (GAA) FET, etc.), embedded memory (e.g., resistive RAM, magnetoresistive RAM, phase change memory (PCRAM), etc.), and the like. It will be appreciated that such a HPC component **106** may be referenced herein as a microprocessor component configured to control operations of the components **102-110** of the semiconductor device **100** in accordance with varying embodiments contemplated herein.

The DTC component **108** may comprise one or more deep trench capacitors used in the operation of the semiconductor device **100**. As indicated in FIG. 1, the semiconductor device **100** also utilizes a printed circuit board component **110** configured to allow integration of the semiconductor device **100** with the final system implementation, e.g., an ultrasound system, display, wand, etc. The skilled artisan will appreciate that the printed circuit board component **110** may provide mechanical (e.g., structural) support, as well as provide electrical routing of the semiconductor device **100** to other devices. The interaction and interconnection of the components **102-110** will be better understood in conjunction with the detailed cross-sectional illustrations provided in FIGS. 2A-2I, discussed below.

Referring now to FIGS. 2A-2I, there are shown partial cross-sectional views of various stages of the manufacturing of the semiconductor device **100**. It will be appreciated that the semiconductor device **100** illustrated in FIGS. 2A-2I is represented as a three-dimensional MEMS package module that comprises a plurality of stacked components, e.g., MEMS devices, analog circuits, microprocessor(s), power components, and circuit board/substrate. It will be understood that the illustration of FIGS. 2A-2I are intended as an example implementation of the system and methods described hereinafter.

In particular, it will be understood that FIGS. 2A-2I illustrate manufacturing stages of the various components (i.e., a MEMS component **102**, an analog circuit component **104**, a high performance chip (HPC) component **106**, a deep trench capacitor (DTC) component **108**, and a circuit board component **110**) used in assembling the semiconductor device **100** of FIGS. 2A-2I. Further, it will be appreciated that the various components **102-110** may be fabricated

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parallelly and independently without specific sequence of manufacturing. That is, the HPC component **106** may be fabricated in a different fab unit and prior to manufacture of the MEMS component **102**, or vice versa. Accordingly, the skilled artisan will appreciate that the illustrations in FIGS. 2A-2I are intended as non-exhaustive example depictions of various stages of manufacture/formation of the components **102-110** utilized in the semiconductor device **100**.

Turning now to FIG. 2A, there is shown a first cross-sectional view of the various components of the semiconductor device **100** during an initial stage of fabrication. As indicated above, each of the components **102-110** may be manufactured separately and independently prior to final formation of the semiconductor device **100**. In FIG. 2A, the MEMS component **102** is depicted as comprising a MEMS membrane substrate **112** and a MEMS sidewall component **114**. As will be appreciated, the MEMS membrane substrate **112** may be constructed of Si, SiO₂, SiN, SiON, or the like, and include a dielectric film coating. The MEMS sidewall component **114** may be constructed of similar materials as that of the MEMS membrane substrate **112**, as will be appreciated by the skilled artisan. In some embodiments, the MEMS sidewall component **114** may be initially formed as a single layer of material on the analog circuit component **104**, and patterned to form cavities **118** between the MEMS sidewall components **114**.

In other embodiments, the cavities **118** and MEMS sidewall components **114** may be patterned from the MEMS membrane substrate **112**. As illustrated in FIG. 1A, the MEMS sidewall components **114** are formed on a MEMS/analog bonding area **116**, disposed an analog circuit insulative layer **120** (of the analog circuit component **104**). Suitable bonding for the bonding area **116** may include, for example and without limitation, eutectic bonding, fusion bonding, hybrid bonding, or the like. In addition to the foregoing, when the MEMS component **102** is implemented as a CMUT or PMUT, the skilled artisan will appreciate that the MEMS component **102** further includes one or more bottom or sensing electrodes (not shown), positioned within the cavity **118** for actuation of the MEMS membrane substrate **112** during operations of the semiconductor device **100**. In accordance with varying embodiments contemplated herein, the aforementioned bottom electrode may comprise, for example and without limitation titanium (Ti) or other metal (e.g., Al, Cu, AlCu, Ag, Au, W, or the like), a metal nitride (e.g., titanium nitride (TiN)), another conductive material, or suitable combinations thereof. The bottom/sensing electrodes may be deposited by, for example and without limitation, CVD, PVD, ALD, electroless plating, electrochemical plating, sputtering, ion metal plasma, another deposition process, or any suitable combination thereof.

The illustration of the semiconductor device **100** of FIG. 2A further includes an analog circuit component **104**, positioned between the MEMS component **102** and the HPC component **106**. As shown in FIG. 2A, the analog circuit component **104**, at this stage of manufacturing, includes an analog circuit insulative layer **120** positioned on an analog circuit substrate **130**. As indicated above, the analog circuit component **104** may include various analog components (e.g., HV pulsers, TIA, MIM, etc.). FIG. 2A illustrates analog complimentary metal oxide semiconductor (CMOS) components **122**, analog CMOS component vias **124**, and analog electrical routing components **126** (configured to function as routing for the analog CMOS components **122**).

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A fusion/eutectic/hybrid bonding area **128** is also illustrated in FIG. 2A associated with the MEMS/analog bonding area **116**.

The semiconductor device **100** of FIG. 2A further includes an HPC component **106** that includes an HPC insulative layer **132** (e.g., a CMOS oxide material) formed on an HPC substrate **144**. As shown in FIG. 2A, the HPC component **106** includes HPC metal components **134** and HPC metal component vias **136**. In accordance with some embodiments, the HPC metal components **134** may be implemented as integrated circuit components, including, for example and without limitation active components (e.g., transistors (the aforementioned FinFET, GAA, etc.), passive components (e.g., capacitors, inductors, resistors, and the like), or combinations thereof. In such embodiments, the HPC metal component vias **136** may be implemented to interconnect the HPC metal components **134**. As will be appreciated, the integration of the HPC component **106** in the semiconductor device **100** integrates advanced CMOS processing capabilities to provide high computing MEMS actuating and sensing advantages. The HPC component **106** may further include a bond pad via **138** for electrical routing to connect the analog component **104** with the HPC component **106** through a bond pad **142**. The bond pad **142** is positioned within a conjunction insulation layer **140** formed on the HPC substrate **144**, as illustrated in FIG. 2A. Additional details of the bond pad via **138** and the bond pad **142** are shown in FIG. 3 below.

In FIG. 3, there is shown a top view and side view of the ordered formation of the bond pad **142**, the bond pad via **138** and positioning thereof relative to the HPC metal components **134** within the HPC insulative layer **132**. As illustrated in FIG. 3, the bond pad via **138** extends through the bond pad **142**, enabling electrical connectivity between the analog circuit component **102** and the HPC metal component **134**. As shown in FIG. 3, stacking of the HPC metal components **134** (i.e., CMOS metals) below the bond pad **142** provides structural support to prevent cracking in the HPC insulative layer **132** (i.e., the CMOX oxide material), as well as providing electromagnetic interference (EMI) shielding. In addition, as shown in FIG. 3, the HPC metal components **134** adjacent to the bond pad **142** may be separated by a distance "D" from the other HPC metal components **134** (shown at **300**) in the cross-sectional view, so as to prevent cracking in the HPC insulative layer **132**. In accordance with some embodiments, the distance "D" should be greater than 1 μm , and within the range of about 1 μm –5 μm . The length "E" of the bond pad via **142** should be greater than 0.6 μm , and within the range of about 0.6 μm –6 μm . In some embodiments, as illustrated in FIG. 3, the bond pad **142** should be situated on the underlying HPC metal component **134**, with a distance "C" from the outside of the bond pad **142** to the edge of the HPC metal component **134** greater than 0.1 μm , and within the range of about 0.1 μm –3 μm . In such an embodiment, the bond pad **142** may have a length "B" of about 2.4 μm , and within the range of 2.4 μm –4.4 μm .

Returning to FIG. 2A, the DTC component **108** of the semiconductor device **100** includes a DTC carrier substrate **146**, one or more DTC carrier substrate metal routings **148**, and one or more DTC dies **150** (e.g., deep trench capacitors). In some embodiments, the DTC dies **150** include a plurality of deep trench capacitors to add capacitance to various integrated circuits, e.g., the various devices of the HPC component **106**. The DTC component **108**, as shown in FIG. 2A, is positioned between the HPC component **106** and the circuit board component **110**. The circuit board component **110** of FIG. 2A is depicted as including a circuit board **152**

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and one or more solder balls **153**. According to some embodiments, the circuit board component **110** is configured to provide a final electrical connection platform for the semiconductor device **100**, and may be implemented as a printed circuit board (PCB), or the like.

Turning now to FIG. 2B, there is illustrated another step in the manufacturing of the semiconductor device **100** in accordance with one embodiment. As shown in FIG. 2B, the analog component substrate **130** and the HPC component substrate **144** are depicted as having been thinned down in preparation of backside through silicon via (TSV) processing. It will be appreciated that such thinning of the analog component substrate **130** and the HPC component substrate **144** may be accomplished via chemical-mechanical polishing (CMP) or other suitable manufacturing method.

In FIG. 2C, the aforementioned TSV processing has been initiated on the analog circuit component **104**, the HPC component **106**, and the DTC component **108**. As shown in FIG. 2C, analog component TSV cavities **154** have been formed in the analog circuit component **104**. That is, an analog component TSV cavity **154** has been etched (e.g., patterned) into the analog substrate **130** in preparation for deposition of TSV material allowing connection between the analog circuit component **104** and the HPC component **106**. In the HPC component **106**, first HPC component TSV cavities **156** have been patterned and formed in the HPC insulative layer **132** between the HPC metal component **134** and through HPC substrate **144**. Second HPC component TSV cavities **158** have been patterned and formed in the HPC substrate **144** in preparation for deposition of TSV material allowing connections between the HPC component **106** and the DTC component **108**. In the DTC component **108**, first DTC component TSV cavities **160** are formed through the DTC carrier substrate **146** in preparation for deposition of TSV material allowing connections between the HPC component **106** and the circuit board component **110**. Second DTC component TSV cavities **162** are formed within the DTC carrier substrate **146** in preparation for deposition of TSV material allowing connections between the HPC component **106** and the DTC carrier substrate metal routing **148**.

Turning now to FIG. 2D, the semiconductor device **100** is now illustrated with TSV material **164** having been deposited in the various TSV cavities **154**–**162**. In accordance with some embodiments, the TSV material **164** may be implemented as a suitable conductive material, including, for example and without limitation, Cu, AlCu, or the like. As shown in FIG. 2D, the analog component TSV **166** (i.e., the analog component TSV cavity **154** filled with the TSV material **164**) may be implemented with a lowest thickness (i.e., the portion through the analog circuit component substrate **130**) may be in the range of about 1 μm –2 μm , and a highest thickness (i.e., the portion contacting the analog electrical routing **126**) may be in the range of about 8 μm –10 μm . The first HPC component TSV **168** (i.e., the first HPC component TSV cavity **156** filled with the TSV material **164**) may be implemented with a lowest thickness (i.e., the portion through the HPC substrate **144**) may be in the range of about 1 μm –2 μm , and a highest thickness (i.e., the portion contacting the HPC metal component **134**) may be in the range of about 8 μm –10 μm . The second HPC component TSV **170** (i.e., the second HPC component TSV cavity **158** filled with the TSV material **164**) may be implemented with a lowest thickness (i.e., the portion through the HPC substrate **144**) may be in the range of about

1 μm –2 μm , and a highest thickness (i.e., the portion contacting the HPC metal component **134**) may be in the range of about 8 μm –10 μm .

The first DTC component TSV **172** (i.e., the first DTC component TSV cavity **160** filled with the TSV material **164**) may be implemented with a lowest thickness (i.e., the portion through the bottom of the DTC carrier substrate **146**) may be in the range of about 1 μm –2 μm , and a highest thickness (i.e., the portion through the top of the DTC carrier substrate **146**) may be in the range of about 8 μm –10 μm . The second DTC component TSV **174** (i.e., the second DTC component TSV cavity **162** filled with the TSV material **164**) may be implemented with a lowest thickness (i.e., the portion through the bottom of the DTC carrier substrate **146**) may be in the range of about 1 μm –2 μm , and a highest thickness (i.e., the portion contacting the carrier substrate metal routing **148**) may be in the range of about 8 μm –10 μm . It will be appreciated that the TSVs **166**–**174** may be deposited by, for example and without limitation, CVD, PVD, ALD, electroless plating, electrochemical plating, sputtering, ion metal plasma, another deposition process, or any suitable combination thereof.

In FIG. 2E, the MEMS membrane substrate **112** is bonded to the MEMS sidewall **114** via any suitable bonding method, e.g., fusion, eutectic, hybrid, etc. Thereafter, as shown in FIG. 2E, the MEMS component **102** is bonded to the analog circuit component **104**. In some embodiments, the MEMS membrane substrate **112** facilitates operation of the MEMS component **102** of the semiconductor device **100** (e.g., the MEMS membrane substrate **112** is the portion of the MEMS component **102** device that oscillates in response to an acoustic wave thereby producing a variable capacitance between a bottom electrode (not shown) and a top electrode (not shown) disposed on the MEMS membrane substrate **112** to produce an electrical signal; or conversely, the MEMS membrane substrate **112** is electrically energized with an AC signal to induce oscillation of the MEMS membrane substrate **112** to generate an acoustic wave). In some embodiments, the MEMS membrane substrate **112** is designed to contact a landing area of the bottom electrode (not shown) on each oscillation, thus providing a binary (i.e., digital) CMUT output. In accordance with some embodiments, the thickness of the MEMS membrane substrate **112** may be thinned after bonding, e.g., utilizing CMP or other mechanisms, which may increase the sensitivity of the MEMS component **102**, durability of the membrane substrate **112**, and the like.

In FIG. 2F, an isolated trench **176** is illustrated in the MEMS component **102**. According to some embodiments, the isolated trench **176** extends through the MEMS membrane substrate **112** and the MEMS sidewall **114** to the MEMS/analog bonding area **116**, thereby providing mechanical isolation between individual MEMS pixels **178**. As used herein, each MEMS pixel **178** corresponds to an individual MEMS device, i.e., that portion of the MEMS component **102** and analog circuit component **104** that are bonded together. In some embodiments, the isolated trench **176** is filled with an isolated material, such as a porous insulated material (e.g., oxide, polyamide, etc.), a composite material, or the like. It will be appreciated that when the isolated trench **176** is not utilized between MEMS pixels **178**, mechanical coupling may occur, i.e., interference between adjoining MEMS pixels **178** during transmission or receiving.

Referring now to FIG. 4, there is shown another embodiment of the isolated trench **176** (of FIGS. 2F–2I) utilizing a second methodology. In FIG. 4, the isolated trench **176** is

patterned at the chip level and an insulated molding **182**, e.g., rubber, polymer, etc., is applied at the final device level to provide mechanical isolation between MEMS pixels **178**. It will be appreciated that the approach described above with respect to FIG. 2F and the approach illustrated in FIG. 4 each provide mechanical isolation between MEMS pixels **178**, thereby insulating the individual membranes of the pixels **178** from being impacting by operations of adjoining pixels **178**. Other benefits to the use of the isolation material in the isolated trench **176** and the isolation molding **182** may include, for example and without limitation, electrical or electro-magnetic interference insulation between pixels **178**.

Returning now to FIG. 2G, there is shown the attachment of the combined MEMS component **102**/analog circuit component **104** to the HPC component **106**. As illustrated in FIG. 2G, the analog component vias **166** are now in contact with the bond pad via **138**. Accordingly, the analog CMOS components **122** are thereby electrically connected to the HPC metal components **134** utilizing the analog component vias **166** and the bond pad vias **138** through the bond pad **142**. It will be appreciated that the bonding of the analog circuit component **104** to the HPC component **106** may be accomplished via any suitable bonding means. Further, each MEMS pixel **178** is now in electrical communication with the HPC component **106**, enabling the HPC component **106** to control operations thereof.

Turning now to FIG. 2H, there is shown the bonding of the DTC component **108** to the HPC component **106**. As illustrated in FIG. 2H, the first HPC component vias **168** are now in contact with the first DTC component vias **172** or in contact with the DTC die **150**, thereby providing connectivity of the HPC metal components **134** to the first DTC component vias **172** or to the capacitors of the DTC die **150**. The second HPC component vias **170** are depicted now in contact with the DTC die **150** through the HPC substrate **144**. The interconnectivity of the first HPC component vias **168** to the first DTC component vias **172** will be better understood in view of FIG. 2I, which illustrates attachment of the DTC component **108** and the circuit board component **110**.

As illustrated in FIG. 2I, the previously joined components (i.e., the MEMS component **102**, the analog circuit component **104**, the HPC component **106**, and the DTC component **108**) are now joined to the circuit board component **110** via the various solder balls **153** shown on the top side of the printed circuit board **152**. As shown in FIG. 2I, the previous joining of the first HPC component vias **168** to the first DTC component vias **172** thereby enable connectivity of the HPC component **106** with the printed circuit board **152**. Further, the second DTC component vias **174** provide connectivity from the DTC carrier substrate metal routing **148** to the printed circuit board **152**.

FIG. 2I further illustrates cascaded seal rings **180** positioned between MEMS pixels **178** to provide noise shielding capability to the semiconductor device **100**, i.e., external noise or noise between MEMS pixels **178**). It will be appreciated that in some embodiments, the cascaded seal ring **180** surrounds every MEMS pixel **178** of a MEMS array (shown in FIGS. 5A–5B as the MEMS array **500**). In the cross-sectional view of FIG. 2I, however, the skilled artisan will appreciate that the analog circuit component **104** includes several metal components or layers (i.e., analog electrical routing **126**), however not all such analog electrical routing **126** may be used as a seal ring, as some of the analog electrical routing **126** is utilized for circuit routing or other purposes. Accordingly, while not all of the analog

electrical routing 126 is used as seal rings, the cascaded seal rings 180 are connected/grounded horizontally/vertically to provide shielding capability.

For example, the top view of a portion of a MEMS array 500 comprising a plurality of MEMS pixels 178 is illustrated in FIG. 5A. A simplified cross-sectional view of the MEMS array 500 is illustrated in FIG. 5B. Referring now to FIG. 5A, each MEMS pixel 178 is illustrated with an exemplary seal ring 180 which surrounds the MEMS pixel 178, thereby providing shielding to the active components located within the seal ring 180. As illustrated in FIG. 5A, the cavity 118 is positioned within the seal ring 180, along with the corresponding sensing electrodes (not shown). For illustrative purposes, the analog component TSV 166 is shown, providing vertical and horizontal connections of the cascaded seal rings 180. As previously discussed, some embodiments contemplated herein provide for the seal ring 180 to surround each MEMS pixel 178, but not fully surrounded at all layers of the MEMS component 102, whereas the circuit layers of the analog circuit component 104 and the HPC component 106 do not. As will be appreciated, the circuit portions of the semiconductor device 100 (shown in FIG. 5B as the analog circuits 502 and the HPC circuits 504) generally have a larger area than a single MEMS pixel 178, so usage of the same pattern of the seal ring 180 in the analog circuit component 104 and the HPC component 106 is problematic. However, in accordance with one embodiment, as illustrated in FIG. 5B, the vertical grounding of the seal ring 180 (i.e., utilizing TSV components 166, 168, and 172) may still be accomplished, thereby providing additional shielding to the MEMS pixels 178.

The skilled artisan will appreciate that the foregoing description of the joining of the various components 102-110 is intended solely as one possible order of forming the semiconductor device 100. Accordingly, any order of joining the components 102-110 is contemplated herein, e.g., joining the circuit board component 110 to the DTC component 108, then attaching the HPC component 106 to the DTC component 108, followed by attaching the analog circuit component 104 to the HPC component 106, and finishing with the bonding of the MEMS component 102 to the analog circuit 104. As another illustrative example, the HPC component 106 may first be attached to the DTC component 108, with the attachment of the MEMS component 102 to the analog circuit component 104 occurring simultaneously therewith (i.e., in another location, fab, facility, chamber, etc.), whereafter the two (the HPC component 106/DTC component 108 combination and the MEMS component 102/analog circuit component 104) are attached together followed by affixing the four combined components (i.e., 102-108) to the circuit board component 110. Accordingly, the skilled artisan will appreciate that the order of attachment presented in FIGS. 2E-2I are intended merely to illustrate the contact points and connections between the components 102-110 and not to limit the order in which the semiconductor device 100 is assembled.

Turning now to FIGS. 6A-6D, there is shown the positioning of through-silicon vias (TSV) in accordance with varying embodiments of the subject application. That is, FIGS. 6A-6D provide alternate embodiments of the positioning of the analog component TSV 166 relative to the cavity 118 and the cascaded seal ring 180. As shown in FIG. 6A, a first analog component TSV 166 pattern is illustrated, wherein the analog component TSVs 166 are positioned at each corner of the MEMS pixel 178, i.e., outside the cavity 118 and/or cascaded seal ring 180 location. In FIG. 6B, the analog component TSVs 166 are located directly below the

cavity 118, i.e., overlapping with the seal ring 180. In FIG. 6C, the analog component TSVs 166 are positioned surrounding the cavity 118, thereby overlapping the seal ring 180. In FIG. 6D, the analog component TSVs 166 are densely proportioned below the cavity 118 and overlapping with the cavity 118 and/or seal ring 180. As shown in FIG. 6D, each analog component TSV 166 may be implemented a distance "A" from the adjoining TSV 166, wherein the distance "A" is greater than 0.5 μm and may be in the range of about 0.5 μm -5 μm . Further, the diameter of the analog component TSV 166 in FIG. 6D may be greater than 0.5 μm and may be in the range of about 0.5 μm -5 μm .

It will be appreciated that the illustration of the seal ring 180 in FIGS. 6A-6D is shown to illustrate the position of the TSVs 166 and that while depicted as a square, the seal ring 180 may be deposited and formed as in a circular manner around the cavity 118 in accordance with various design considerations. Accordingly, the illustrations in FIGS. 6A-6D are intended merely to illustrate variations in the formation of the analog component TSV 166 positioning relative to the location of the active components, i.e., the sensing electrodes, movable membrane, etc. Accordingly, FIGS. 7A-7D illustrate formation of the seal ring 180 in a circular formation. As shown in FIG. 7A, a first analog component TSV 166 pattern is illustrated, wherein the analog component TSVs 166 are positioned at each corner of the MEMS pixel 178, i.e., outside the cavity 118 and/or cascaded seal ring 180 location. FIG. 8 provides an isometric illustration of an individual MEMS pixel 178 in accordance with the embodiment set forth in FIG. 7A. As shown in FIG. 8, the MEMS pixel 178 may have a length in the range of about 50 μm -1000 μm .

Returning to FIG. 7B, a second embodiment of a circular seal ring 180 pattern is depicted, wherein the analog component TSVs 166 are located directly below the cavity 118, i.e., overlapping with the seal ring 180. In FIG. 7C, the analog component TSVs 166 are positioned surrounding the cavity 118, thereby overlapping the seal ring 180. In FIG. 7D, the analog component TSVs 166 are densely proportioned below the cavity 118 and overlapping with the cavity 118 and/or seal ring 180. As shown in FIG. 7D, each analog component TSV 166 may be implemented a distance "A" from the adjoining TSV 166, wherein the distance "A" is greater than 0.5 μm and may be in the range of about 0.5 μm -5 μm . Further, the diameter of the analog component TSV 166 in FIG. 7D may be greater than 0.5 μm and may be in the range of about 0.5 μm -5 μm .

Referring now to FIG. 9, there is shown a flowchart illustrating a method for forming the semiconductor device 100 in accordance with some embodiments. As shown in FIG. 9, the method 900 begins with the fabrication of the various components 102-110 of the semiconductor device 100. As indicated above, the fabrication of each component 102-110 may occur independently and in any order of fabrication. In some embodiments, one or more of the components 102-110 may be fabricated at different facilities. Further, in other embodiments, one or more components 102-110 may be fabricated at different times prior to formation of the semiconductor device 100. That is, the MEMS component 102 may be fabricated at a first fab (Fab 1), the analog component 104 may be fabricated at a second fab (Fab 2), the HPC component 106 may be fabricated at a third fab (Fab 3), the DTC component 108 may be fabricated at a fourth fab (Fab 4), and the circuit board component 110 may be fabricated at a fifth fab (Fab 5). Further, operations at any of the aforementioned fabs (Fabs 1-5) may occur simultaneously, sequentially, or variations thereof. It will be

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appreciated that each component **102-110** may utilize different manufacturing techniques and technologies. Thus, fabrication of the components **102-110** may occur at the aforementioned different fabs. Accordingly, at **902**, initial fabrication of the components **102-110** is performed. For example, the MEMS component **102** is fabricated in Fab 1, the analog circuit component **104** is fabricated in Fab 2, the HPC component **106** is fabricated in Fab 3, the DTC component **108** is fabricated at Fab 4, and the circuit board component **110** is fabricated at Fab 5.

At **904**, the analog circuit component **104** is processed to reduce the thickness of the analog circuit component substrate **130**. In some embodiments, CMP or other means are used to reduce the thickness of the analog circuit component substrate **130** in anticipation of the formation of TSV cavities **154**. At **906**, the HPC component **106** is processed to reduce the thickness of the HPC substrate **144**. In some embodiments, CMP or other means are used to reduce the thickness of the HPC substrate **144** in anticipation of the formation of TSV cavities **156** and **158**. It will be appreciated that the processing of steps **904** and **906** may be performed at the initial fabs, or alternatively may be performed at different fab.

At **908**, at least one TSV cavity **154-162** is formed in the analog circuit component **104**, the HPC component **106**, and the DTC component **108**. As will be appreciated, the formation of the TSV cavities **154-162** may be performed at the fab corresponding to the initial fabrication of the components **104-108**, at one of the initial fabs, or at a different fab.

At **910**, TSV electrical connections are formed in the TSV cavities **154-162** of the analog circuit component **104**, the HPC component **106**, and the DTC component **108**. That is, TSV material **164** is deposited in the analog component TSV cavity **154** to form the TSV electrical connection, i.e., the analog component TSV **166**. Similarly, TSV material **164** is deposited in the first HPC component TSV cavity **156** and the second HPC component TSV cavity **158** to form the TSV electrical connections, i.e., the first HPC component TSV **168** and the second HPC component TSV **170**, respectively. In addition, TSV material **164** is deposited in the first DTC component TSV cavity **160** and the second DTC component TSV cavity **162** to form the TSV electrical connections, i.e., the first DTC component TSV **172** and the second DTC component TSV **174**, respectively. As will be appreciated, the formation of the TSV electrical connections, i.e., the TSVs **166-174** may be performed at the fab corresponding to the initial fabrication of the components **104-108**, at one of the initial fabs, or at a different fab. In some embodiments, the TSV material **164** may be implemented as AlCu, Cu, Al, or other suitable conductive metal.

As indicated above, the bonding of the components **102-110** to each other may be performed in any suitable order, and the steps illustrated in FIG. 9 are intended solely to indicate the bonding positions and not the order of when the components are bonded to each other. That is, steps **912**, **920**, **922**, and **924** may be performed in any order and the illustration in FIG. 9 is intended as one possible order for bonding the components **102-110**. Accordingly, at **912**, the MEMS membrane substrate **112** of the MEMS component **102** is bonded to the top of the analog circuit component **104**, thereby forming an integrated MEMS component **102/** analog circuit component **104**. In accordance with some embodiments, the bonding may be accomplished via fusion bonding, eutectic bonding, hybrid bonding, etc. It will be appreciated that the bonding described herein corresponds to wafer-level bonding. At **914**, the MEMS membrane substrate **112** is processed to reduce the thickness thereof.

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Suitable mechanisms for thickness reduction include, for example and without limitation, CMP, etching, etc. At **916**, isolated trenches **176** are patterned on the MEMS membrane substrate **112**. It will be appreciated that patterning of the isolated trenches **176** may employ any suitable patterning technique such as a photolithographic patterning technique using deposition of a photoresist layer and selective exposure via a photomask to visible light, ultraviolet light, deep ultraviolet light (i.e., DUV lithography), extreme ultraviolet light (i.e. EUV lithography), or so forth, followed by development of the exposed photoresist and subsequent etching to remove the uncovered portions of the MEMS membrane substrate **112**. In other embodiments, patterning of an electron-sensitive resist layer may be by way of electron beam exposure (electron beam lithography, i.e., e-beam lithography). The skilled artisan will appreciate that the foregoing are merely illustrative examples.

At **918**, the isolated trenches **176** are filled with an isolated material, such as a porous insulated material (e.g., oxide, polyamide, etc.), a composite material, or the like. The isolated material may be deposited by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), electroless plating, electrochemical plating, sputtering, ion metal plasma, another deposition process, or any suitable combination thereof.

At **920**, the integrated MEMS component **102/** analog circuit component **104** is bonded to the HPC component **106**, thereby forming an integrated MEMS component **102/** analog circuit component **104/** HPC component **106**. That is, the bottom of the analog circuit component **104** is bonded to the top of the HPC component **106**. Suitable bonding methodologies include, for example and without limitation, fusion bonding, eutectic bonding, hybrid bonding, or the like. At **922**, the integrated MEMS component **102/** analog circuit component **104/** HPC component **106** is bonded to the DTC component **108**, thereby forming an integrated MEMS component **102/** analog circuit component **104/** HPC component **106/** DTC component **108**. That is, the bottom of the HPC component **106** is bonded to the top of the DTC component **108**. At **924**, the integrated MEMS component **102/** analog circuit component **104/** HPC component **106/** DTC component **108** is bonded to the circuit board component **110** to complete formation of a semiconductor device **100** in accordance with some embodiments contemplated herein. At **924**, the bottom of the DTC component **108** is bonded (e.g., attached) to the top of the circuit board component **110**, e.g., using the solder balls **153** to electrically connect the printed circuit board **152** to the DTC component TSVs **172-174**. It will be appreciated that the foregoing processing, occurring at the wafer level, may subsequently be sent for packaging, i.e., dicing the wafer into individual MEMS arrays. Thereafter, assembly into various articles of manufacture that utilize such MEMS arrays, i.e., fingerprint scanners, ultrasound devices, etc., may occur.

Turning now to FIG. 10, there is shown a functional block diagram **1000** of one example operation of the semiconductor device **100** described above. As shown in FIG. 10, an acoustic signal is transmitted by the MEMS component **102** of the semiconductor device **100** at **1002**. In the embodiment of FIG. 10, the MEMS component **102** may be implemented as a CMUT array, a PMUT array, or the like. For example, as illustrated in FIG. 11, the semiconductor device **100** may be disposed within a handheld ultrasound wand **1100**, that includes the semiconductor device **100**, wand body **1102**, and rubber barrier **1104**. As illustrated in FIG. 11, the

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ultrasound wand **1100** is placed in contact with skin **1108**, with a dielectric grease **1106** positioned between the rubber barrier **1104** and skin **1108**. At **1004**, the acoustic signal then penetrates the skin **1108**. At **1006**, the acoustic signal returning from within the skin **1108** is received by MEMS component **102** and translated into an electric signal. This electric signal is then transmitted to the analog circuit component **104**, e.g., to a transimpedance amplifier at **1008** and thereafter converted into a digital signal that is communicated to the HPC component **106**. At **1010**, the HPC component **106** processes the signal (e.g., artificial intelligence, DSP, etc.) and outputs the results of such processing to a designated output terminal (not shown) such as, for example and without limitation, a video monitor, a portable electronic device (e.g., tablet, laptop, mobile phone, etc.), a remote computer (e.g., via wireless or wired communications), or the like.

In accordance with a first embodiment, there is provided a method of forming a semiconductor device. The method includes processing an analog circuit component to reduce the thickness of an analog circuit component substrate, and processing a high performance chip (HPC) component to reduce the thickness of an HPC component substrate. The method further includes forming at least one through-silicon via (TSV) cavity in the analog circuit component, the HPC component, and the deep trench capacitor (DTC) component. Additionally, the method includes forming a TSV electrical connection on each of the analog circuit component, the HPC component, and the DTC component in the at least one TSV cavity, and bonding a microelectromechanical system (MEMS) membrane substrate of a MEMS component to a top of the analog circuit component to form an integrated MEMS component/analog circuit component. The method further includes bonding the bottom of the analog circuit component of the integrated MEMS component/analog circuit component to the top of the HPC component to form an integrated MEMS component/analog circuit component/HPC component, and bonding the bottom of the HPC component of the integrated MEMS component/analog circuit component/HPC component to the top of the DTC component to form an integrated MEMS component/analog circuit component/HPC component/DTC component. The method also includes bonding the bottom of the DTC component of the integrated MEMS component/analog circuit component/HPC component/DTC component to the top of a circuit board component.

In accordance with a second embodiment, there is provided a semiconductor device that includes a MEMS component that has a MEMS membrane substrate and a MEMS sidewall, such that the MEMS component includes a plurality of MEMS pixels. The semiconductor device also includes an analog circuit component that is bonded to the MEMS component, and which includes at least one analog CMOS component within an analog circuit insulative layer, and an analog circuit component substrate. The semiconductor device further includes an HPC component that is bonded to the analog circuit component substrate. The HPC component includes at least one HPC metal component disposed within an HPC insulative layer, at least one bond pad, at least one bond pad via connecting the at least one bond pad and the at least one HPC metal component, and an HPC substrate. In addition, the semiconductor device includes a DTC component that is bonded to the HPC substrate. The DTC component includes at least one DTC die disposed in a DTC carrier substrate.

In accordance with a third embodiment, there is provided a method of forming a semiconductor device that includes

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fabricating a MEMS component, an analog circuit component, an HPC component, and a DTC component. The method also includes bonding a microelectromechanical system (MEMS) membrane substrate of the MEMS component to a top of the analog circuit component to form an integrated MEMS component/analog circuit component, and bonding a bottom of the analog circuit component of the integrated MEMS component/analog circuit component to a top of the HPC component to form an integrated MEMS component/analog circuit component/HPC component. The method further includes bonding a bottom of the HPC component of the integrated MEMS component/analog circuit component/HPC component to a top of the DTC component to form an integrated MEMS component/analog circuit component/HPC component/DTC component. Furthermore, the method includes bonding a bottom of the DTC component of the integrated MEMS component/analog circuit component/HPC component/DTC component to a top of a circuit board component.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of forming a semiconductor device, comprising:

- processing an analog circuit component to reduce a thickness of an analog circuit component substrate;
- processing a high performance chip (HPC) component to reduce a thickness of an HPC component substrate;
- forming at least one through-silicon via (TSV) cavity in the analog circuit component, the HPC component, and a deep trench capacitor (DTC) component;
- forming, in the at least one TSV cavity, a TSV electrical connection on each of the analog circuit component, the HPC component, and the DTC component;
- bonding a microelectromechanical system (MEMS) membrane substrate of a MEMS component to a top of the analog circuit component to form an integrated MEMS component/analog circuit component;
- bonding a bottom of the analog circuit component of the integrated MEMS component/analog circuit component to a top of the HPC component to form an integrated MEMS component/analog circuit component/HPC component;
- bonding a bottom of the HPC component of the integrated MEMS component/analog circuit component/HPC component to a top of the DTC component to form an integrated MEMS component/analog circuit component/HPC component/DTC component; and
- bonding a bottom of the DTC component of the integrated MEMS component/analog circuit component/HPC component/DTC component to a top of a circuit board component.

2. The method of claim 1, further comprising:
processing the MEMS membrane substrate to reduce a thickness thereof;

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patterning at least one isolated trench on the MEMS membrane substrate; and
filling the at least one isolated trench on the MEMS membrane substrate with an isolating material.

3. The method of claim 2, wherein the MEMS component comprises at least one of a capacitive micromachined ultrasonic transducer or a piezoelectric micromachined ultrasonic transducer.

4. The method of claim 1, further comprising forming at least one cascaded seal ring in the MEMS component.

5. The method of claim 1, wherein the bonding of the MEMS component to the analog circuit component is performed by fusion bonding, eutectic bonding, or hybrid bonding.

6. The method of claim 1, wherein the bonding of the integrated MEMS component/analog circuit component to the HPC component is performed by fusion bonding, eutectic bonding, or hybrid bonding.

7. The method of claim 1, wherein the bonding of the integrated MEMS component/analog circuit component/HPC component to the DTC component is performed by fusion bonding, eutectic bonding, or hybrid bonding.

8. The method of claim 1, wherein forming the at least one TSV cavity further comprises:

forming an analog component TSV cavity on the analog circuit component;

forming a first HPC component TSV cavity and a second HPC component TSV cavity on the HPC component; and

forming a first DTC component TSV cavity and a second DTC component TSV cavity on the DTC component.

9. The method of claim 8, further comprising:

forming an analog circuit component TSV in the analog circuit component TSV cavity;

forming a first HPC component TSV in the first HPC component TSV cavity and a second HPC component TSV in the second HPC component TSV cavity on the HPC component; and

forming a first DTC component TSV in the first DTC component TSV cavity and a second DTC component TSV in the second DTC component TSV cavity on the DTC component,

wherein the analog circuit component TSV is in contact with an HPC metal component of the HPC component, the first HPC component TSV is in contact with the first DTC component TSV, the second HPC component is in contact with at least one DTC on the DTC component, and the first DTC component TSV and the second DTC component TSV are in contact with the circuit board component.

10. A method of forming a semiconductor device, comprising:

fabricating a microelectromechanical system (MEMS) component, an analog circuit component, a high performance chip (HPC) component, and a deep trench capacitor (DTC) component;

bonding a MEMS membrane substrate of the MEMS component to a top of the analog circuit component to form an integrated MEMS component/analog circuit component;

bonding a bottom of the analog circuit component of the integrated MEMS component/analog circuit component to a top of the HPC component to form an integrated MEMS component/analog circuit component/HPC component;

bonding a bottom of the HPC component of the integrated MEMS component/analog circuit component/HPC

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component to a top of the DTC component to form an integrated MEMS component/analog circuit component/HPC component/DTC component; and

bonding a bottom of the DTC component of the integrated MEMS component/analog circuit component/HPC component/DTC component to a top of a circuit board component.

11. The method of claim 10, further comprising forming at least one cascaded seal ring in the MEMS component.

12. The method of claim 11, further comprising:

processing the MEMS membrane substrate to reduce a thickness thereof;

patterning at least one isolated trench on the MEMS membrane substrate; and

filling the at least one isolated trench on the MEMS membrane substrate with an isolating material.

13. A method of forming a semiconductor device, comprising:

forming a microelectromechanical system (MEMS) component comprising a MEMS membrane substrate and a MEMS sidewall, the MEMS component including a plurality of MEMS pixels;

forming an analog circuit component bonded to the MEMS component, the analog circuit component comprising at least one analog complementary metal oxide (CMOS) component within an analog circuit insulative layer, and an analog circuit component substrate;

bonding a high performance chip (HPC) component to the analog circuit component substrate, the HPC component comprising at least one HPC metal component disposed within an HPC insulative layer, at least one bond pad, at least one bond pad via connecting the at least one bond pad and the at least one HPC metal component, and an HPC substrate; and

bonding a deep trench capacitor (DTC) component to the HPC substrate, the DTC component comprising at least one DTC die disposed in a DTC carrier substrate.

14. The method of claim 13, further comprising disposing at least one cascaded seal ring in the MEMS component to provide shielding for noise between adjoining MEMS pixels.

15. The method of claim 14, wherein forming the analog circuit component further comprises forming at least one analog component through-silicon via (TSV) in contact with the at least one analog CMOS component and positioned in the analog circuit substrate.

16. The method of claim 15, further comprising: forming the at least one analog component TSV in contact with the at least one bond pad of the HPC component.

17. The method of claim 16, wherein forming the HPC component further comprises forming a first HPC component TSV in contact with the at least one HPC metal component and forming a second HPC component TSV disposed in the HPC substrate.

18. The method of claim 17, wherein the DTC component further comprises a first DTC component TSV disposed in the DTC carrier substrate, and a second DTC component TSV in contact with the at least one DTC die.

19. The method of claim 18, wherein the MEMS membrane substrate further comprises at least one isolated trench positioned between adjoining MEMS pixels.

20. The method of claim 19, wherein the at least one isolated trench includes an isolation material, the isolation material providing mechanical interference shielding between the adjoining MEMS pixels.

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